From Fields to Dose: Project Record

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Internal end-to-end record of the study, from concept to the current state, for the author's reference.

1. Purpose and how to use this record

This record collects, in one place and in chronological and thematic order, everything the author needs to recall about the study "From Fields to Dose: Evaluating Large Language Models for Weight-Based Pediatric Medication Extraction and Dosing Consistency": the question and the constraints adopted at the outset, the registration, the build of the synthetic test set and its lock, the model runs and the provider events that interrupted them, every analysis performed and its key outputs, the findings as they now stand and how four rounds of external adversarial review changed their wording, the integrity controls and the one discrepancy they surfaced, the official document set, the operating commands that remain to be run, the pending work, and the decisions and anticipated challenges with prepared answers. It is written for the author, not for an outlet or a reviewer; the publishable text is the feature article, the authoritative account of every number is the study report, the software description is the system architecture document, and the reviewer's entry point is the review guide. Use this record to re-enter the project after an interval: read Section 2 for the study in one page, Section 3 for the dated sequence of events, Section 11 for the commands still to be run, and Section 12 for the release checklist. Every number here is copied from the study report, the review guide, the system architecture document or the recorded facts of the project journey; where a fact is absent from those sources the record says "not recorded" rather than supplying one. Tables and verbatim passages are copied rather than paraphrased so that this record does not become a third version of any figure.

2. The study in one page

The question. Weight-based pediatric dosing turns a clinical note into a dose through several dependent steps: identify the current weight, the weight-normalised dose, its basis (per day or per dose), the frequency and the route, then compute the amount per administration. A language model that extracts each element correctly can still imply the wrong dose, and a model that abstains correctly on one missing element can still supply an unsupported value for another. The study measures those outcomes directly.

Research questions (verbatim from Protocol v1.0, Section 1, as reproduced in the study report).

RQ1 (field level). How accurately do contemporary LLMs extract each element of a weight-based pediatric medication regimen from narrative synthetic records?

RQ2 (regimen level). How often are all required elements correct together, and how large is the gap between mean field accuracy and complete-regimen accuracy?

RQ3 (dosing consistency and consequence). When extracted values are used to recompute mg/kg/dose and mg/kg/day, how often does the result differ from the intended regimen by more than a pre-specified tolerance, and which error classes produce dose-consequential errors?

RQ4 (abstention). When required information is absent or conflicting, do models abstain appropriately or supply unsupported values?

RQ5 (architecture). Does a hybrid workflow (LLM interpretation plus deterministic arithmetic) reduce dose-consequential errors relative to LLM end-to-end reconstruction?

RQ6 (moderators, exploratory). How do documentation complexity and pediatric age band affect the above?

Constraints adopted at the outset and kept throughout. No real patient data; no clinician or pharmacist review; no employer data, systems, name or resources; API-only access to pinned model identifiers; a 200 USD spending ceiling for the test phase; all materials releasable. The study is solo and self-funded. The ground truth is known by construction, because each record is generated from a canonical regimen, so correctness does not depend on human adjudication; plausibility is enforced by automated gates (label ranges, growth-reference weights, age floors), an automated realism audit, and a blinded author read of a 10 percent sample whose correction rate is reported. Synthetic data gives full releasability, no protected information, and control of the exact failure paths under test (stale weight, discontinued order, second medication, pound-denominated weight, missing interval); the cost is generalisation, and the study bounds performance on documentation it can fully specify. Both points are stated in the protocol.

Registration. Open Science Framework open-ended registration https://osf.io/3p9ey, DOI 10.17605/OSF.IO/3P9EY, project https://osf.io/uajs3. Registered 3 October 2026 before any evaluated model saw a test record. The registration carries the protocol, the model configuration and the SHA-256 hashes of the test set and of the scoring code.

Test set. 361 synthetic pediatric records (ages birth to 18 years; weights sampled from WHO and CDC growth references; regimens for 21 medications taken from FDA labeling; three documentation complexity levels; 57 abstention cases; a 108-record stratified repeat subset), locked on 3 October 2026 (commit 528c2b5).

Models. Five, under the identifiers as used: gpt-5.5-2026-04-23, claude-sonnet-5, gemini-3.1-pro-preview (OpenAI-compatible endpoint), deepseek-v4-pro, llama3.1:8b (Ollama, local; digest 46e0c10c039e). Paraphrase generator and realism auditor: qwen2.5:14b (local), a family distinct from every evaluated model; the harness refuses an evaluated model that shares a family with the generator.

Workflows. Workflow B: the model extracts all fields and computes dose_per_admin_mg, reporting its own mg_per_kg_per_dose and mg_per_kg_per_day. Workflow C: the model extracts the fields only; the deterministic calculator computes exposure at scoring time. Workflow A (a rules baseline) was not run, a logged deviation decided before results.

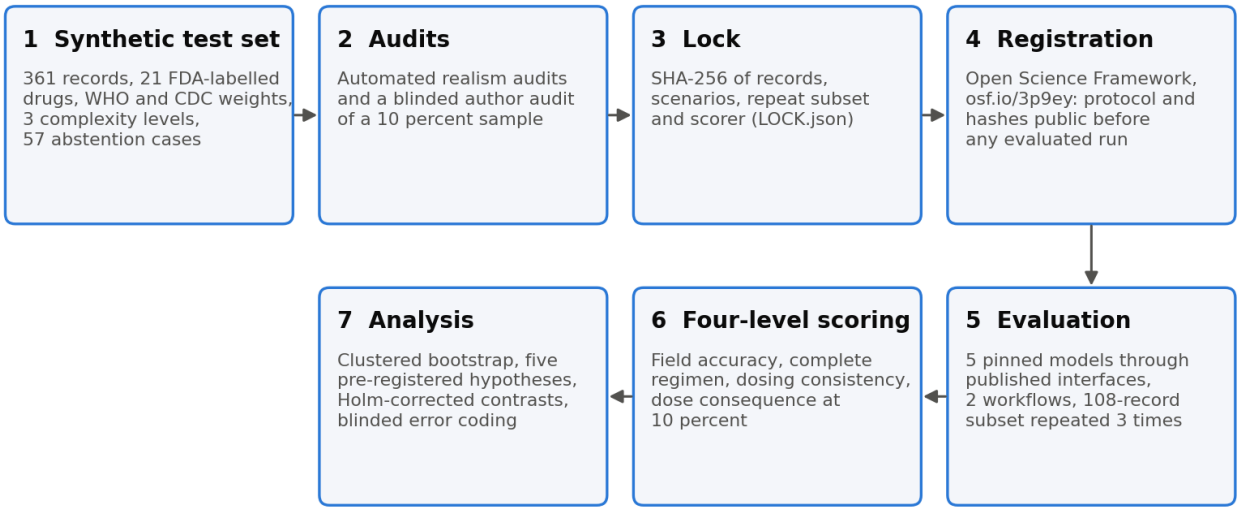


Figure 1. The study in seven steps.

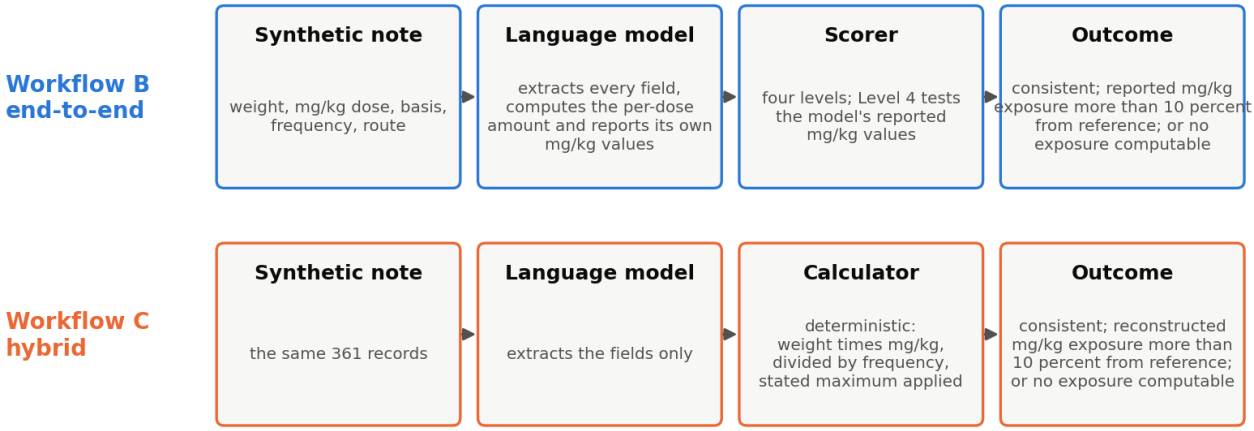


Figure 2. The two workflows and the composite outcome.

Scoring. Four levels. Level 1: field accuracy after normalisation (1 percent numeric tolerance). Level 2: complete-regimen accuracy; dose_per_admin_mg is not required when the note withholds it. Level 3: dosing consistency (observed mg/kg/dose and mg/kg/day within tolerance of canonical). Level 4: composite dose-consequence; dose_consequential is True when the worse of the two relative deviations exceeds 10 percent or when no exposure can be computed (either mg/kg value missing); dose_consequential_sens applies 20 percent. The two components, "computed exposure more than 10 percent off" and "no exposure computable", are reported separately wherever the composite is reported. The 10 percent figure is adapted from a published modified-Delphi definition of pediatric dosing error; it is an operational threshold on a constructed intended dose, not a harm claim. exceeds_labeled_max is a separate flag that tests the observed daily exposure against max_per_day_mg only. Primary endpoint: complete dose-consistent regimen accuracy on non-abstention records; a malformed output counts as a failure. Abstention cases are scored as appropriate abstention, unnecessary abstention and unsupported-value hallucination as implemented.

Pre-specified hypotheses (verbatim from the study report).

H1. Complete-regimen accuracy will be at least 10 percentage points lower than mean field accuracy for every evaluated model.

H2. The dose-consequential error rate will increase monotonically from complexity Level 1 to Level 3.

H3. The hybrid workflow (C) will have a lower dose-consequential error rate than the end-to-end workflow (B) for the same model (primary comparison).

H4. Dose-basis confusion (mg/kg/day vs mg/kg/dose) and weight-selection errors will together account for the largest share of dose-consequential errors.

H5 (exploratory). Performance will be lower in the neonate and infant bands than in the adolescent band under matched challenges.

Statistics (Protocol Section 16). Unit of analysis record; clustering by scenario; clustered bootstrap, 2,000 resamples. H3 paired at the record level per model; supported if the CI excludes zero in the hypothesised direction for a majority of models. Secondary pairwise model contrasts with Holm correction (implemented after registration). Exploratory mixed-effects regression not performed.

Current state. Four of five models complete on every run; Gemini 3.1 Pro has workflow C run1 complete, workflow B run1 at 342 of 361, and no repeat runs; completion is planned at the next quota reset. Spend under 100 USD against the 200 USD ceiling.

3. Timeline

Calendar date	What happened	Evidence
2 October 2026	dosing_sources.csv built from FDA labels for 21 medications; growth reference files added; label tools written; a both-bases rule (a drug carrying both per-dose and per-day rows, gentamicin) downgraded to a warning and logged in DEVIATIONS.md	Commit b249a75; sources/dosing_sources.csv; protocol/DEVIATIONS.md
3 October 2026, early	Development set of 172 records; prompt finalised with an example output object after two prompt defects on development data (no output schema; JSON-Schema type descriptors copied literally); scorer fixes; billing-error handling	Commit 7c948c4, the last change to scoring/score.py; data/dev/
3 October 2026	Test set built; first automated realism audit found three generator defects, fixed; second audit flagged 45 of 363 records, all reviewed, no generator error; two neonatal Level 3 records with pre-birth prior dates removed; blinded author self-audit of 36 records with no corrections; test set locked: 361 records, 57 abstention cases, 108-record repeat subset	Commit 528c2b5; data/test/LOCK.json (locked_at 2026-10-03); data/test/records.jsonl, scenarios.jsonl, repeat_subset.txt
3 October 2026, later	Protocol v1.0 finalised (ORCID added, limitations and AI-assistance disclosure, Section 18 reduced to one line, PDF metadata reduced to title and author) and registered on OSF; registration recorded in protocol/README.md	https://osf.io/3p9ey; protocol/README.md; protocol/LOCK_hashes.txt

4 October 2026	run1 on the test set for GPT-5.5 and Claude Sonnet 5 (both workflows), DeepSeek V4 Pro and Llama 3.1 8B (local); repeat runs run2 and run3 on the 108-record subset for those four models	results/ raw_test_<workflow>_<model>_<run>.jsonl with their _config lines; run manifest (first and last call timestamps per file, 4 to 5 October 2026)
4 October 2026	Gemini 3.1 Pro workflow C run1 complete; workflow B run1 reached 342 of 361 before the per-project daily quota (250 requests per model per Google Cloud project per day on the paid tier) stopped it; a second Google Cloud project under the same billing account set up and the key swap in .env documented	DEVIATIONS.md row dated 4 October (Gemini requests split across two projects); results/raw_test_B_gemini-3.1-pro-preview_run1.jsonl
4 October 2026	DeepSeek returned two persistently empty responses in run2 (three attempts), scored malformed	DEVIATIONS.md row dated 4 October
4 October 2026	Provider credit exhaustion stopped runs at times (OpenAI, Anthropic, DeepSeek); runs resumed after top-ups; resumption re-queries only records without parsed output	DEVIATIONS.md row dated 4 October (resumed runs re-query records with no parsed output); runner behaviour in harness/runner.py
4 October 2026	Workflow A (rules baseline) not run, decided before results	DEVIATIONS.md row dated 4 October
5 October 2026	Scoring of all completed run files; four-model analysis (primary, stability, pairwise, per-drug)	results/analysis_primary_four_models.txt; results/analysis_stability_four_models.txt; results/analysis_pairwise_four_models.txt; results/per_drug_run1.csv
5 October 2026	Three external adversarial reviews of the feature article processed and a fourth review of the official set; row-level resolution of every frontier composite failure	Review guide, review history tables; study report 5.10 and 5.11; results/packet/02_row_level/
5 October 2026	Written-amount, stratified and maximum-check tools added outside the locked set	tools/admin_amount.py, tools/by_stratum.py, tools/max_check.py; results/admin_amount_run1.csv, by_stratum_run1.csv, max_check_run1.csv; DEVIATIONS.md rows dated 5 October
5 October 2026	Lock verification completed, including the discovery that the malformed-row correction had been applied inside the locked analysis script, its restoration byte for byte and the move of the correction to analysis/analyze_malformed_as_failure.py, after which hash_scoring() reproduced the registered value 7be9bc8704e09b94014741847f16b047b115c7b2c7c7846ec01afe03445461ed	results/logs/ analyze_malformed_as_failure.diff (15 lines); results/packet/01_lock_verification.txt; results/analysis_primary_locked.txt and results/analysis_primary_corrected.txt
5 October 2026	Probe of a constructed output through the locked scorer on the neonatal rounding record; probe files removed; Claude scored file verified unchanged by hash	Study report 5.11; DEVIATIONS.md row dated 5 October
5 October 2026	Evidence packet built from the repository (31 files); commit 8776fe6 pushed; official documents issued without development version labels	results/packet/ (six parts); commit 8776fe6
Planned, next quota reset	Gemini 3.1 Pro completion: workflow B run1 remainder plus repeat runs run2 and run3 in both workflows, about 450 calls, split across the two projects	Pending; see Section 11

4. Build phase

4.1 Sources. The dosing table sources/dosing_sources.csv was built on 2 October 2026 from FDA labeling obtained through openFDA: 22 verified rows for 21 drugs, with label set identifiers and SPL versions, dosing bases, ranges and maxima. fetch_labels.py pulls one label per drug from openFDA (query overrides for salts such as amoxicillin and clavulanate potassium). The author verified each row against the label text using show_dosing.py and set_row.py; the notes column records which maxima are stated by the label and which are the author's derivations. check_sources.py enforces completeness and flags a drug carrying both per-dose and per-day rows (gentamicin; accepted and logged, the both-bases rule downgraded to a warning in commit b249a75). The 21 medications: amoxicillin, amoxicillin-clavulanate, cephalexin, cefdinir, azithromycin, clindamycin, fluconazole, ampicillin, ceftriaxone, gentamicin (per-dose and per-day regimens), acetaminophen IV, ibuprofen, ondansetron, dexamethasone, prednisolone, famotidine, levothyroxine, digoxin, fentanyl, midazolam, vancomycin.

Growth references: WHO Child Growth Standards weight-for-age 0 to 24 months (boys and girls files from cdc.gov/growthcharts/who-data-files.htm, merged by hand into sources/wtageinf.csv, 50 rows) and CDC Growth Charts weight-for-age 2 to 20 years (sources/wtage.csv). Weight is sampled by age and sex with z in minus 2 to plus 2, 10 percent tails at z 2.05 to 2.5, and a cap at 2.2 times the median. Age bands (sources/age_bands.json, NICHD terminology): neonate 0 to 27 days; infant 28 to 365 days; toddler and young child 366 to 2190 days, which collapses the NICHD categories toddler (13 months to 2 years) and early childhood (2 to 5 years); school-age child 2191 to 4380 days (NICHD middle childhood); adolescent 4381 to 6574 days.

4.2 Generation. build_corpus.py samples age within band, sex, and weight from the LMS tables, chooses a drug permitted for the age, draws a dose within the label range, and renders a note from templates.py at the requested complexity. Complexity levels: Level 1, a bare order with weight, drug, dose, frequency and route; Level 2, an HPI and plan with vitals, a second medication (cholecalciferol under 6 months, cetirizine otherwise) and a computed per-dose amount with volume; Level 3, an admission note adding a prior-visit weight with its date, a discontinued order for the same drug, and in some records weights in pounds. Of the 103 non-abstention Level 3 scenarios, 36 withhold the per-dose amount (challenge tag derive_per_dose), so the model must derive it; 67 print it. Abstention cases (57): a required element is absent ("interval to be confirmed") or only a dated prior weight is present; the correct response is null with the field named in insufficient_information. A fraction of records is paraphrased by qwen2.5:14b and audited by the same model with a committed rubric (audit.py); flagged records are reviewed by the author.

The note prints the per-dose amount as an order would, with practical rounding and with labelled maxima applied; the scenario's expected_dose_per_admin_mg is that printed amount. Audit of the 304 non-abstention scenarios (caps applied, units matched): about 53 print an amount more than 1 percent from the exact product and one (neonatal dexamethasone, 0.02 mg printed for an exact 0.0226 mg, a 13 percent gap) is more than 10 percent from it. The consequences of that convention for scoring are in Section 6.

Development phase (172 records, data/dev/). Two prompt defects were found and fixed on development data: no output schema (models invented key names); JSON-Schema type descriptors copied literally by one model. The final prompt shows a concrete example object (amoxicillin values), the abstention rule, and no rule on the permissible age of a documented weight. Scorer fixes before lock: non-scalar values treated as missing;

dose_per_admin_mg not required at Level 2 when the note withholds it; billing and quota errors stop the runner. The last change to scoring/score.py is commit 7c948c4 (3 October 2026, before the lock).

4.3 Audits. The first automated realism audit flagged three generator defects (unbounded tail weights to 108 kg; prior-visit dates before birth; cetirizine in young infants); all were fixed and the set rebuilt. The second audit flagged 45 of 363 records; all were reviewed; no generator error. Two Level 3 neonatal records with pre-birth prior dates were removed and the generator corrected, giving 361 records (two variants per scenario after audit removals). Blinded author self-audit of 36 records: no note requiring correction; three reader entry slips, reported.

4.4 Lock. data/test/LOCK.json was written by tools/lock_testset.py on 3 October 2026 (commit 528c2b5). It holds the record count, SHA-256 of records.jsonl, scenarios.jsonl and repeat_subset.txt as raw file bytes, and a scoring hash computed by hash_scoring() as one SHA-256 over the concatenated bytes of seven files: scoring/score.py, generator/calculator.py, harness/prompts/system.txt, harness/prompts/task_B.txt, harness/prompts/task_C.txt, harness/output_schema.json and analysis/analyze.py. The lock function, verbatim from tools/lock_testset.py:

```
def hash_file(p):

    h = hashlib.sha256(); h.update(open(p, "rb").read()); return h.hexdigest()

def hash_scoring():

    h = hashlib.sha256()

    for p in ("scoring/score.py", "generator/calculator.py", "harness/prompts/system.txt",
"harness/prompts/task_B.txt", "harness/prompts/task_C.txt", "harness/output_schema.json",
"analysis/analyze.py"):

        h.update(open(os.path.join(ROOT, p), "rb").read())

    return h.hexdigest()
```

LOCK.json content: locked_at 2026-10-03; n_records 361; smoke false; and the four hashes:

Item	SHA-256
records_sha256 (data/test/records.jsonl)	492d7ca2209719eb8bf96548403dab1e26cb9fed36be3482733dd69790cea7f4
scenarios_sha256 (data/test/scenarios.jsonl)	432be74fe65c2164ad9e21705821f5814055994f389526614aa744a671c220cb
repeat_subset_sha256 (data/test/repeat_subset.txt)	f1012bdcb01f9cdcf80cf000b0ad6831352d9a98a6e57a0e8ebd207dbdfa4196
scoring_sha256 (hash_scoring() over the seven files)	7be9bc8704e09b94014741847f16b047b115c7b2c7c7846ec01afe03445461ed

The runner refuses to start a test run without the lock file; scoring/score.py check_lock(split) loads data/test/LOCK.json and exits if hash_file(records.jsonl) or hash_scoring() differs. The hashes are in protocol/LOCK_hashes.txt and in the OSF registration.

4.5 Protocol and registration. Later on 3 October 2026, Protocol v1.0 was finalised: ORCID added, limitations and AI-assistance disclosure added, Section 18 reduced to one line, PDF metadata reduced to title and author. It was registered on OSF as an open-ended registration (https://osf.io/3p9ey, DOI 10.17605/OSF.IO/3P9EY; project https://osf.io/uajs3) before any evaluated model saw a test record, and the registration was recorded in

protocol/README.md. The registration carries the protocol, the model configuration and the hashes above, so a reader can verify later that neither the test set nor the scoring code was changed after the fact. The protocol folder holds Protocol v1.0 (docx and PDF), LOCK_hashes.txt, DEVIATIONS.md, README.md and an archive. Protocol Section 20 covers AI assistance: assistants as development and drafting aids under the author's direction; not evaluated models; an overlapping family evaluated only through its pinned API.

5. Run phase

The protocol registered three workflows: a rules-only Workflow A with no language model, which was not run (recorded in the deviations log, with the reason that it lies outside the hypotheses and that the calculator in the hybrid workflow already provides the non-model arithmetic reference), and Workflows B and C, which keep their registered letters because the letters are fixed in the registration, in the locked prompt files and in every result file; renumbering them after registration would make the results disagree with the registered design.

5.1 Models and workflows. Identifiers as used: gpt-5.5-2026-04-23, claude-sonnet-5, gemini-3.1-pro-preview (OpenAI-compatible endpoint), deepseek-v4-pro, llama3.1:8b (Ollama, local; digest 46e0c10c039e). Every provider is accessed through the same OpenAI-compatible client; the only per-provider differences are endpoint and key name. config/models.yaml holds evaluated models, generator, auditor, budget ceiling and repeat settings, with environment-variable names only; provider keys live in a git-ignored .env. The GPT-5.5 identifier carries a release date; the Claude and Gemini identifiers name a release; the DeepSeek identifier resolves to the provider's current release at call time and is therefore not immutable, so the manifest records first and last call timestamps per run file (all test runs 4 to 5 October 2026) rather than claiming every provider version was fixed. Each model sees every record under workflow B (extract all fields and compute dose_per_admin_mg; report mg_per_kg_per_dose and mg_per_kg_per_day) and under workflow C (extract the fields only).

5.2 run1 and repeats. On 4 October 2026 run1 was executed on the full test set (361 records) for GPT-5.5 and Claude Sonnet 5 in both workflows, and for DeepSeek V4 Pro and Llama 3.1 8B (local), followed by repeat runs run2 and run3 on the 108-record stratified repeat subset for those four models in both workflows. The full design is 30 run files: five models, two workflows, three runs. Each raw file (results/raw_test_<workflow>_<model>_<run>.jsonl) opens with a _config line carrying model id, endpoint, workflow, run id, subset, started, system_prompt_sha256 and task_prompt_sha256; the prompt hash observed had the 70e815de3c01 prefix for all runs. Then one line per attempt: record id, scenario id, workflow, run id, raw text, parsed object or null, error or null, retries, latency, timestamp. Malformed outputs after resumption: 0 of 361 for every completed model in run1.

5.3 Resumption semantics. The runner marks a record done only when a parsed object exists and re-queries the rest on restart; a non-JSON response triggers one repair message; a still-unparseable response after a second failure is stored with parsed null; transient errors are retried; billing, credit, quota and rate-limit errors stop that model's loop with a message to rerun later. A record may therefore appear more than once in a raw file if a run was resumed; the scorer keeps the last attempt. Resumed runs give unparsed outputs an extra attempt, applied uniformly to all models and reported (before and after malformed counts). This is a logged deviation dated 4 October. Runs are resumable and idempotent at the record level; any interrupted command is rerun unchanged to resume. One background job runs per chain (caffeinate plus nohup); progress is read from the output files by counting records with parsed output.

5.4 Quota and credit events. Provider credit exhaustion stopped runs at times for OpenAI, Anthropic and DeepSeek, and the runs were resumed after top-ups. OpenAI and Anthropic prepaid credit exhaustion stops the runner with a 429 or 400; DeepSeek returns 402 on insufficient balance. DeepSeek returned two persistently empty responses in workflow B run2 on three attempts; they are scored malformed under the protocol rule (deviation dated 4 October). Gemini 3.1 Pro is subject to 250 requests per model per Google Cloud project per day on the paid tier; the quota stopped workflow B run1 at 342 of 361 after workflow C run1 had completed (361). A second Google Cloud project under the same billing account was set up, with the same model identifier and parameters (logged), and the key swap in .env documented: both keys are kept in .env, one active (deviation dated 4 October: Gemini requests split across two Google Cloud projects).

5.5 Gemini status and the plan to complete it. Status: workflow C run1 complete (361); workflow B run1 at 342 of 361; repeat runs run2 and run3 not started in either workflow. Plan: at the next quota reset, run the workflow B run1 remainder plus run2 and run3 in both workflows, about 450 calls, split across the two projects using the Gemini chain and key-swap commands in Section 11; then score, rerun the analyses, regenerate Figures 3, 5 and 6 with the Gemini row, and compute H3 under both wordings for five models. The study report's rationale for keeping Gemini: it is one of the five models in the public registration; removing it after more than 700 outputs exist is the deviation a reviewer reads as selective reporting, with "quota" as the only available reason; partial reporting would unbalance the strata and the paired comparison; keeping it costs about 15 USD and one day; and H3's verdict under Section 1 no longer depends on Gemini. Grok was proposed as a sixth model and declined: not pre-registered, a post hoc addition, almost certainly at ceiling, about 1,150 calls plus coding time without changing a conclusion.

5.6 Costs. Approximate test-phase spend from provider dashboards: OpenAI about 37 USD, Google about 28 USD, Anthropic about 13 USD, DeepSeek about 11 USD; Llama local at no provider cost. Under 100 USD against the 200 USD ceiling. Observed per-call cost: GPT-5.5 about 3.1 cents, Gemini about 2.2 cents, Claude about 1.0 cent, DeepSeek about 0.9 cents. Final figures from provider dashboards go in the run manifest.

6. Analysis phase

6.1 Scoring (scoring/score.py, locked). score.py reads each raw file, keeps the last attempt per record, joins the scenario, normalises units, compares fields at 1 percent tolerance (NUM_TOL = 0.01), scores abstention cases, and assigns automatic error classes by rules. REQUIRED = ["drug", "current_weight_kg", "dose_basis", "weight_normalized_dose", "dose_unit", "dose_per_admin_mg", "frequency_per_day", "route"]; dose_per_admin_mg is dropped from the required set when the scenario carries the derive_per_dose challenge. The targets object sets dose_per_admin_mg from the scenario's expected_dose_per_admin_mg. A malformed output (parsed null) takes an early branch with dose_consequential True and codes ["malformed_output"]. The dose_cap automatic code fires when the scenario has a maximum, the output reports none, and the per-dose amount is wrong or missing. tol is 0.10 and tol_sens is 0.20. The Level 3 and 4 path, verbatim:

```
req = [k for k in REQUIRED if not (k == "dose_per_admin_mg" and "derive_per_dose" in
(s.get("challenges") or []))]

fields = {k: field_correct(k, p.get(k), t[k]) for k in req}

out["fields"] = fields

out["field_acc"] = sum(fields.values()) / len(fields)

out["complete_regimen"] = all(fields.values())
```

```
out["codes"] = auto_code(p, s, t)

if abst:

    flagged = set(p.get("insufficient_information") or [])

    expect = {"current_weight_kg" if f == "current_weight_value" else f for f in miss}

    out["appropriate_abstention"] = expect.issubset(flagged) and all(p.get(f) is None for f in
expect)

    out["unnecessary_abstention"] = bool(flagged - expect)

    out["hallucinated"] = any(p.get(f) is not None for f in expect)

    out["dosing_consistent"] = None; out["dose_consequential"] = None

    return out

# Level 3/4

if workflow == "C":

    try:

        rc = reconstruct(p.get("current_weight_kg"), p.get("dose_basis"),
p.get("weight_normalized_dose"), p.get("dose_unit"),

                         p.get("frequency_per_day"), p.get("max_per_dose_mg"),
p.get("max_per_day_mg"), p.get("concentration_mg_per_ml"))

        obs_dose, obs_day = rc["mg_per_kg_per_dose"], rc["mg_per_kg_per_day"]

    except Exception:

        obs_dose, obs_day = None, None

else:

    obs_dose, obs_day = p.get("mg_per_kg_per_dose"), p.get("mg_per_kg_per_day")

exp_dose, exp_day = s.get("expected_mg_per_kg_per_dose"), s.get("expected_mg_per_kg_per_day")

dev_dose, dev_day = relative_deviation(obs_dose, exp_dose), relative_deviation(obs_day, exp_day)

devs = [d for d in (dev_dose, dev_day) if d is not None]

worst = max(devs) if devs else None

out.update({"obs_mg_per_kg_per_dose": obs_dose, "obs_mg_per_kg_per_day": obs_day,

            "dev_dose": dev_dose, "dev_day": dev_day,

            "dosing_consistent": (worst is not None and worst <= tol),

            "dose_consequential": (worst is None or worst > tol),

            "dose_consequential_sens": (worst is None or worst > tol_sens),

            "exceeds_labeled_max": bool(s.get("max_per_day_mg") and obs_day is not None and obs_day
* to_kg(s["current_weight_value"], s["current_weight_unit"]) > s["max_per_day_mg"] * 1.001)})
```

```
if out["complete_regimen"] and not out["dosing_consistent"]:
out["codes"].append("arithmetic_inconsistency")

out["primary_endpoint"] = out["complete_regimen"] and out["dosing_consistent"]

return out
```

Two consequences of this path and of the printed-amount convention, recorded in the study report. (a) In workflow B the observed mg/kg values are the fields the model itself reports, so a model that derives those from a copied printed amount inherits its rounding; a model that reports the exact product rather than the printed amount is scored incomplete at Level 2 and correct at Levels 3 and 4, which is what happened to GPT-5.5 in nine workflow-B records. (b) On the one neonatal record no single internally consistent per-dose amount satisfies both the 1 percent Level 2 tolerance on the printed amount (0.0198 to 0.0202 mg) and the 10 percent Level 4 tolerance on the mg/kg figure read back from it (equivalent to 0.0203 to 0.0249 mg). Because workflow B scores dose_per_admin_mg at Level 2 and the model's reported mg/kg fields at Level 4 with no consistency check between them, an output reporting 0.02 mg in the amount field and 0.005 mg/kg per dose in the computed field passes both; this was verified by running that constructed output through the locked scorer; no model produced such an output. The written per-dose amount (dose_per_admin_mg) enters Level 2 at a 1 percent tolerance and the exploratory check of 6.9; it does not enter Level 4 in either workflow. Each scored row (results/scored_test_<workflow>_<model>_<run>.jsonl) carries record id, scenario id, workflow, run id, age band, complexity, abstention flag, conflict flag, challenges, generation method, malformed flag, per-field correctness, field accuracy, complete regimen, automatic codes, observed mg/kg per dose and per day, deviations, dosing consistent, dose consequential (10 percent) and sensitivity (20 percent), exceeds labelled maximum (per-day maxima only), primary endpoint.

6.2 Registered primary analysis (analysis/analyze.py, locked; results/analysis_primary_four_models.txt). analyze.py takes the run1 scored files and reports, per model and workflow: primary endpoint, field accuracy, complete regimen, composite failure rate with clustered bootstrap CIs (2,000 resamples over scenarios); abstention metrics; breakdowns by complexity, band and generation method; malformed counts; top error codes; and the H3 paired comparison on the composite rate with its CI and verdict. analysis/analyze_malformed_as_failure.py is the same script with malformed non-abstention rows counted as failures on primary endpoint, complete regimen and dosing consistency (previously dropped from the bootstrap denominator); it is used for any file containing malformed rows (the two DeepSeek run2 records) and gives identical output on run1, where there are no malformed rows (results/analysis_primary_locked.txt and results/analysis_primary_corrected.txt differ only in the print order of one line). Primary endpoint and field accuracy, non-abstention records, n = 304 per model and workflow, as copied from the study report:

Model	Workflow	Field accuracy	Complete regimen	Composite failure	Primary endpoint
GPT-5.5	B	0.996	0.970 (0.947 to 0.987)	0.000	0.970 (0.947 to 0.987)
GPT-5.5	C	1.000	1.000	0.000	1.000
Claude Sonnet 5	B	1.000	0.997 (0.990 to 1.000)	0.016 (0.003 to 0.033)	0.980 (0.964 to 0.993)
Claude Sonnet 5	C	1.000	0.997 (0.990 to 1.000)	0.000	0.997 (0.990 to 1.000)
DeepSeek V4 Pro	B	1.000	0.997 (0.990 to 1.000)	0.007 (0.000 to 0.016)	0.990 (0.977 to 1.000)
DeepSeek V4 Pro	C	1.000	1.000	0.000	1.000
Gemini 3.1 Pro	B	pending	pending	pending	pending

Gemini 3.1 Pro	C	pending	pending	pending	pending
Llama 3.1 8B	B	0.968	0.809 (0.766 to 0.855)	0.645 (0.592 to 0.697)	0.329 (0.276 to 0.382)
Llama 3.1 8B	C	0.759	0.062 (0.036 to 0.089)	0.891 (0.855 to 0.924)	0.056 (0.030 to 0.082)

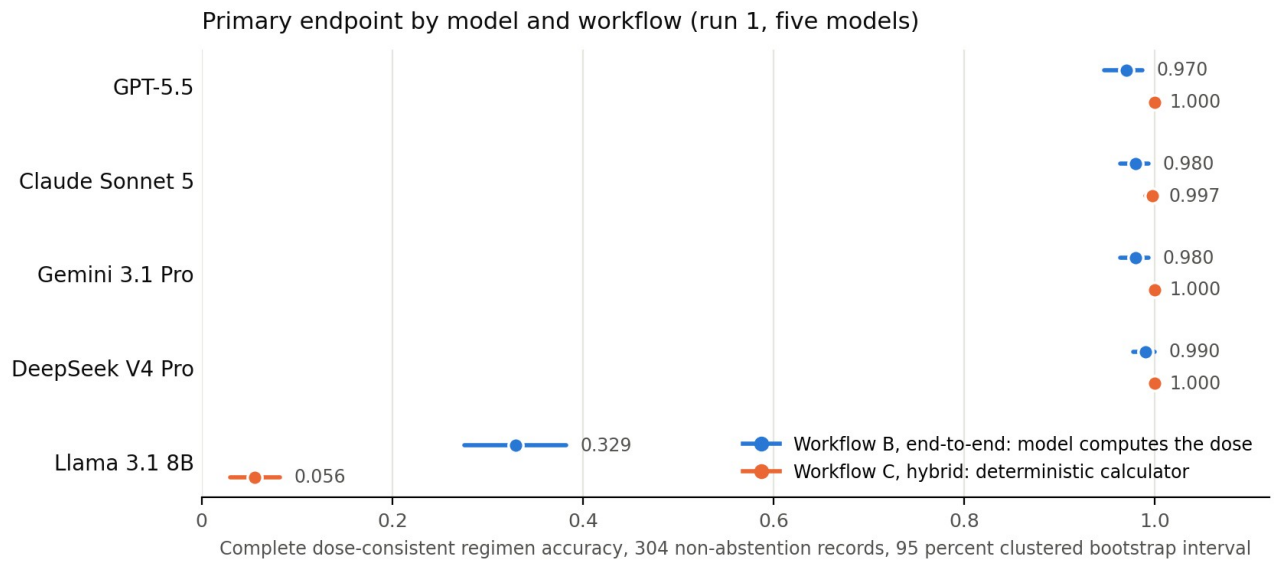


Figure 3. Primary endpoint by model and workflow with 95 percent clustered bootstrap intervals.

Composite failure counts behind the rates (304 records): Claude B 5 (four maximum-dose outputs and the neonatal rounding record); DeepSeek B 2 (both maximum-dose outputs); GPT-5.5 both workflows, Claude C and DeepSeek C 0; Llama B 196, of which 189 reported exposures more than 10 percent off and 7 not computable; Llama C 271, of which 2 and 269. The small model's written per-dose amount, which Level 4 does not use, was within 10 percent of the order in 250 (B) and 249 (C) of 304.

Hypothesis ledger (analysis version: analyze.py with the malformed-row correction of 5 October; run1 files; four models), copied from the study report:

Hypothesis	Pre-specified rule	Result	Verdict
H1	Field accuracy minus complete-regimen accuracy at least 0.10 for every model	GPT-5.5 0.026 (B), 0.000 (C); Claude 0.003, 0.003; DeepSeek 0.003, 0.000; Llama 0.159, 0.697	Not supported (holds for Llama only)
H2	Composite failure rate rises monotonically L1 to L3	Frontier near zero at all levels; Llama B 0.634, 0.700, 0.602; Llama C 0.851, 0.940, 0.883	Not supported
H3 (primary)	C minus B composite failure rate below zero with CI excluding zero, for a majority of models	GPT-5.5 +0.000 (null at ceiling); Claude -0.016 (-0.033 to -0.003), supported; DeepSeek -0.007 (-0.016 to +0.000), not supported; Llama +0.247 (+0.178 to +0.312), reversed; Gemini pending	Not supported: at most 2 of 5 can meet the rule
H4	Dose-basis and weight-selection classes are the largest	Largest automatic classes: adult fixed-dose assumption and	Not supported as automatically coded; manual coding will

	share of composite failures	arithmetic inconsistency (Llama); dose basis distant third; weight selection rare. Row inspection: of the 154 Llama B records with a correct written amount and a failed composite, 74 carry the mg amount in the per-dose mg/kg field, 48 the per-day value in the per-dose field, 15 a null, 17 other; much of "arithmetic inconsistency" is unit and basis confusion inside the reported fields	settle the class
H5 (exploratory)	Neonate and infant bands lower than adolescent	Frontier: no band effect; Llama B 0.571 adolescent, 0.638 infant, 1.000 neonate; Llama C 0.929 adolescent, 0.914 infant, 0.462 neonate	Inconclusive; neonate estimates oppose across workflows

H3 under both protocol wordings. Section 1 (verbatim): "H3. The hybrid workflow (C) will have a lower dose-consequential error rate than the end-to-end workflow (B) for the same model (primary comparison)." Section 16 (verbatim): "Primary comparison (H3): Workflow C minus Workflow B on the primary endpoint, per model, paired at the record level, with clustered bootstrap 95% CI. The comparison is declared supported if the CI excludes zero in the hypothesized direction for the majority of evaluated models." The two name different outcome measures. The analysis follows Section 1; the dated record of that choice is the registration itself: analysis/analyze.py, which implements H3 on the composite rate, is one of the seven files whose combined hash was published in the registration (lock commit 528c2b5, before any test run), so the analysed definition was fixed before any result existed. Section 16's rule (paired, clustered, majority) applies to either measure. The Section 16 comparison is computed by analysis/pairwise.py with its load and paired_boot functions.

Model	C minus B, composite failure rate (Section 1)	C minus B, primary endpoint (Section 16)
GPT-5.5	+0.000 (null at ceiling)	+0.030 (+0.013 to +0.049), supported; the entire gain is the nine Level 2 rounding records, no dose changed
Claude Sonnet 5	-0.016 (-0.033 to -0.003), supported	+0.016 (+0.003 to +0.033), supported
DeepSeek V4 Pro	-0.007 (-0.016 to +0.000), not supported	+0.010 (+0.000 to +0.023), not supported
Gemini 3.1 Pro	pending	pending
Llama 3.1 8B	+0.247 (+0.178 to +0.312), reversed	-0.273 (-0.336 to -0.211), reversed
Verdict	Under Section 1: not supported, at most 2 of 5	Under Section 16: two of four completed; supported only if Gemini also shows the effect (3 of 5)

6.3 Secondary pairwise contrasts (analysis/pairwise.py, added after registration; results/analysis_pairwise_four_models.txt). Primary endpoint, paired by record, clustered bootstrap, Holm-adjusted. No statistically significant pairwise difference among the frontier models was detected in either workflow, which is not evidence of equivalence: Claude minus DeepSeek B -0.010, adjusted p 0.58; Claude minus GPT-5.5 B +0.010, adjusted p 1.00; DeepSeek minus GPT-5.5 B +0.020, adjusted p 0.46; all C contrasts within 0.003, adjusted p 1.00. Every frontier model exceeds Llama: B +0.641 to +0.661; C +0.941 to +0.944; all adjusted p below 0.0001. The exploratory mixed-effects regression was not performed (degenerate at ceiling).

6.4 GPT-5.5 workflow B shortfall (row-level). Nine records fail only on dose_per_admin_mg with no error class and no composite failure: the model reported the exact product (6.15 kg times 2 mg/kg is 12.3 mg) where the order printed a rounded amount (12 mg). The neonatal dexamethasone record is one of the nine.

6.5 Abstention (analyze.py; n = 57 records). Frontier models: 52 of 57 appropriate (0.912, CI 0.842 to 0.982), identical across models and workflows because the same five records account for every miss. The five records are five distinct scenarios: test-clindamycin-mg_kg_day-infant-L3-v0, test-gentamicin-mg_kg_day-school_age-L3-v1, test-dexamethasone-mg_kg_day-toddler-L3-v1, test-dexamethasone-mg_kg_day-school_age-L3-v1, test-dexamethasone-mg_kg_day-adolescent-L3-v0. Each note documents only a weight dated at a prior visit (2025-12-01) against an order dated 2026-03-10; every frontier model supplied that weight as current. For the 8-month-old clindamycin record the prior weight is about 20 percent below the generator's withheld reference weight; for the 13-year-old the gap is marginal. Seven of the 52 appropriate abstentions also withheld a computable per-dose amount (flagged dose cap by the automatic rule; on inspection, a withheld computable field). Llama: 0.772 (B), 0.825 (C).

6.6 Error classes (run1 code counts from analyze.py) and the calculator mechanism. Llama C: extraction dose wrong 270, adult fixed-dose assumption 261, frequency 25, arithmetic 2. Llama B: arithmetic inconsistency 146, frequency 27, extraction dose wrong 17, dose basis 14, weight wrong 1. Claude B: arithmetic 5, dose basis 1. Claude C: dose basis 1. DeepSeek B: arithmetic 2. DeepSeek C and GPT-5.5: none. For the small model the calculator raised the composite failure count from 196 to 271 of 304 and changed its composition: 189 wrong computed exposures and 7 unreconstructable under B became 2 and 269 under C. From the row-level trace: in a typical Llama workflow-C failure the note states "28 mg/kg/day ... (266 mg per dose)"; the model returns dose_basis "fixed", weight_normalised_dose null, dose_per_admin_mg 266; the calculator cannot form an mg/kg exposure from a null weight-normalised dose, the record is scored dosing-inconsistent, and the automatic classes are adult fixed-dose assumption and extraction dose wrong. Complete traces for two B and two C failures are in the evidence packet.

Reconciliation of the composite with the written amount (a cross-tab of admin_amount_run1.csv status against the scored composite and its component; Llama, run1, 304 records per workflow), copied from the study report:

Workflow	Written amount	Composite pass	Reported or computed exposure more than 10 percent off	No exposure computable
B	within 10 percent (250)	96	139	15
B	off (27)	4	21	2
B	missing (27)	8	0	19
C	within 10 percent (249)	14	2	233
C	off (21)	0	0	21
C	missing (34)	19	0	15

Reading, as the study report states it. In B, 154 records have a correct written amount and a failed composite; the model's reported mg/kg fields, not its amount, fail: 74 records carry the milligram amount in the per-dose mg/kg field (note "195 mg per dose", reported mg_per_kg_per_dose 195 or similar), 48 carry the per-day value in the per-dose field (43 mg/kg/day reported as 43 mg/kg/dose), 15 are null, 17 other. In C, 233 records have a correct written amount and no computable exposure because the model labelled the order "fixed" and nulled the mg/kg value. The composite is a regimen-reconstruction failure; measured on the amount it wrote, the small

model matched the order in about five records of six under both workflows; measured on the structured regimen a calculator or range check would use, it failed 196 and 271. A user of workflow B who read only dose_per_admin_mg would see a correct number in 250 of 304 records; a system that consumed the mg/kg fields would see a wrong or empty regimen in 196. The eight records in B where the written amount is missing yet the composite passes are records whose reported mg/kg values were right while the amount field was left empty.

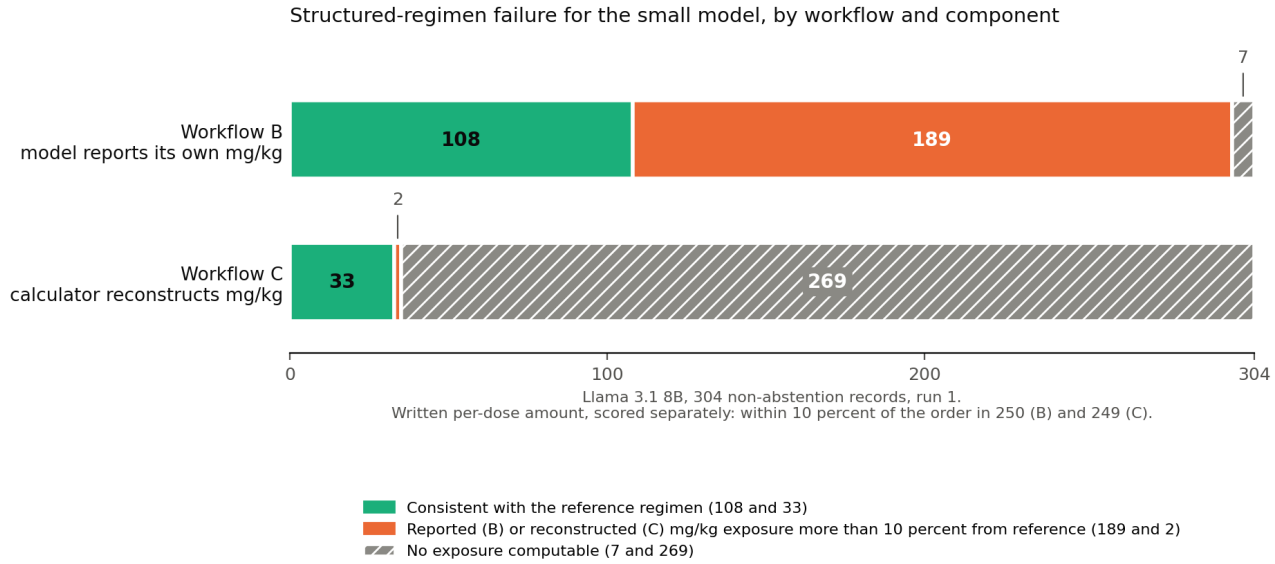


Figure 4. Composite failure for the small model by workflow and component.

6.7 Stability (analysis/stability.py; results/analysis_stability_four_models.txt; 108-record repeat subset, three runs). Agreement 0.972 to 1.000 for all four models in both workflows. Llama at 1.000 reflects deterministic local inference, not robustness. Lowest frontier values: Claude C complete regimen 0.972; GPT-5.5 B dose per administration 0.981; DeepSeek B 0.991 on all fields. Two DeepSeek B run2 records returned empty responses on three attempts and are malformed.

6.8 Exploratory moderators (analyze.py breakdowns; tools/per_drug.py). Paraphrased notes scored lower than template notes in workflow B for every frontier model (Claude 0.948 versus 1.000; GPT-5.5 0.957 versus 0.979; DeepSeek 0.974 versus 1.000). All seven frontier composite failures occurred in paraphrased records; the template-note composite failure rate is 0.000 for every frontier model in both workflows. The four maximum-dose notes state the maximum unambiguously in their paraphrased form ("max 875 mg/dose", "maximum daily dose not to exceed 2000 mg", "not to exceed 60 mg/day"), so paraphrase did not remove the instruction; whether paraphrase raises the failure rate is a hypothesis for a larger set, not a result of this one. Per-drug table: results/per_drug_run1.csv (160 rows).

6.9 Maximum-dose outputs (row-level resolution by a short script that joins a record's note text, scenario fields, the model's last parsed output and its scored row). Five non-abstention Level 3 notes give a weight-based rule, a maximum written into the order, and no per-dose amount, and in each the weight-based product exceeds the maximum. Seven workflow-B outputs wrote the uncapped product: six exceed the maximum by more than 10 percent and are composite failures; the seventh (Claude, prednisolone adolescent v0, 64.65 mg against 60 mg

per day, 7.75 percent over) is inside the 10 percent tolerance and passed Level 4 (dev_dose 0.0775, dose_consequential False, primary endpoint True); the registered exceeds_labeled_max flag is True for it, so the registered per-day flag did record it, and the post hoc maximum check of 6.12 is what brought it into the count of uncapped outputs. In every one of the seven the model had placed the maximum in the correct structured field (max_per_dose_mg or max_per_day_mg) before computing the amount without it; the failure is therefore neither extraction nor a specification gap. GPT-5.5 applied the maximum in all five notes. The automatic scorer labelled the six larger outputs as arithmetic inconsistency; the pattern was identified by row inspection.

Record (scenario)	Order as written	Maximum (field; FDA label set id)	Reference per dose	Claude Sonnet 5	DeepSeek V4 Pro	exceeds_labeled_max
test-amoxicillin-mg_kg_day-school_age-L3-v1-r0 (11 y F, 70.4 kg)	42 mg/kg/day PO divided q12h; max 875 mg/dose	875 mg per dose (max_per_dose_mg; 06e5c07d-d95c-4955-8d35-a4703ca2359e)	875 mg	1478.4 mg	1478.4 mg	False (flag tests per-day maxima only)
test-ceftriaxone-mg_kg_day-adolescent-L3-v1-r0 (17 y M, 153 lb)	54 mg/kg/day IV divided BID; maximum daily dose not to exceed 2000 mg	2000 mg per day (max_per_day_mg; 53df951b-2eee-44b1-bae2-df7ed85bc0ed)	1000 mg	1873.8 mg twice daily (3747.6 mg/day)	1873.8 mg twice daily	True
test-prednisolone-mg_kg_day-school_age-L3-v1-r0 (9 y F, 90.6 lb)	2 mg/kg/day PO q24h; not to exceed 60 mg/day	60 mg per day (max_per_day_mg; 286a6032-41de-435a-b6e6-14052b66b85d)	60 mg	82.19 mg once daily	applied	True
test-prednisolone-mg_kg_day-adolescent-L3-v1-r0 (17 y M, 87.3 kg)	1.5 mg/kg/day PO q24h; not to exceed 60 mg/day	60 mg per day (max_per_day_mg; 286a6032-41de-435a-b6e6-14052b66b85d)	60 mg	130.95 mg once daily	applied	True
test-prednisolone-mg_kg_day-adolescent-L3-v0-r0 (adolescent, 43.1 kg)	1.5 mg/kg/day q24h; not to exceed 60 mg/day	60 mg per day (max_per_day_mg; 286a6032-41de-435a-b6e6-14052b66b85d)	60 mg	64.65 mg once daily (7.75 percent over; within tolerance, Level 4 pass)	applied	True (B); the calculator applied the cap in C, flag False

Provenance of the maxima. The maxima are the ones the synthetic orders state, and the model's task was to apply the maximum written in the order it read. Their derivation from the labels, recorded in sources/dosing_sources.csv, differs by drug. Ceftriaxone: the label states 50 to 75 mg/kg/day once daily or divided every 12 hours, not to exceed 2 g per day, for serious non-meningitis infections; the 2000 mg per day maximum is stated by the label, and the 1000 mg reference per dose is that daily limit divided by the twice-daily frequency. Amoxicillin: the label lists 875 mg every 12 hours as the regimen for patients over 40 kg; the study uses that adult regimen dose as the per-dose ceiling for a child whose weight-based product exceeds it, a common prescribing convention and the author's derivation, not a maximum the label states as such. Prednisolone: the label gives an initial pediatric range of 0.14 to 2 mg/kg/day and cites NHLBI asthma dosing of 1 to 2 mg/kg/day; the 60 mg per day figure is the upper end of the label's general initial-dose range applied as a

ceiling, the author's derivation, not a ceiling the label attaches to asthma. The article therefore describes all four as maxima stated in the order and names only ceftriaxone's as the label's own. No pharmacist reviewed the notes; this is the study's standing constraint and is stated as a limitation.

6.10 The neonatal rounding record (row-level resolution and probe). test-dexamethasone-mg_kg_day-neonate-L1-v0-r0: a 17-day-old, 4.52 kg, dexamethasone 0.02 mg/kg/day PO divided four times daily, note prints "0.02 mg per dose"; exact per-dose 0.0226 mg. Claude Sonnet 5 (B) wrote dose_per_admin_mg 0.02 and mg_per_kg_per_dose 0.0044; the scorer's observed mg/kg/dose is 0.0044 against 0.005 expected, dev_dose 0.12, dev_day 0.0, dosing_consistent False, dose_consequential True, dose_consequential_sens False, exceeds_labeled_max False. GPT-5.5 (B) wrote 0.0226 mg: all fields correct except dose_per_admin_mg (13 percent from the printed 0.02 at a 1 percent tolerance), dev_dose 0.0, dose_consequential False, primary endpoint False at Level 2. DeepSeek V4 Pro (B) likewise wrote 0.0226 mg. The error is in neither extraction nor cap application nor scorer arithmetic; it is a test-set record whose printed rounding exceeds the Level 4 tolerance, on which no internally consistent per-dose amount meets both locked tolerances. Probe (5 October): a constructed output identical to Claude's but with dose_per_admin_mg 0.02, mg_per_kg_per_dose 0.005 and mg_per_kg_per_day 0.02 was written to a probe raw file and scored by the locked scorer: field accuracy 1.0, complete regimen True, dev_dose 0.0, dosing_consistent True, dose_consequential False, primary endpoint True. The scorer therefore does not reject an internally inconsistent output on this record; an earlier draft's claim that no output could pass both levels was too strong and is withdrawn. The probe files were removed and the Claude scored file verified unchanged by hash. The record is scored as registered, disclosed, removed by the 20 percent sensitivity; a Level 2 sensitivity against the exact product is listed as follow-up.

6.11 Written per-dose amount (exploratory; added after registration; tools/admin_amount.py; results/admin_amount_run1.csv and .txt). Question: is the number the model wrote in dose_per_admin_mg usable on its own, independent of the mg/kg fields that Level 4 uses? Rule set: reference is the scenario's expected_dose_per_admin_mg (printed, rounded, maximum applied, mg); "off" is absolute relative deviation above 10 percent; "missing" is absent, null or non-numeric; denominator 304 non-abstention run1 records per model and workflow; no unit conversion (both sides mg). Missing values are split by whether the note withholds the amount (36 Level 3 scenarios) or prints it.

Model	Workflow	Compared	Off (more than 10 percent)	Missing, amount withheld in note	Missing, amount printed
GPT-5.5	B	304	1 (neonatal rounding record)	0	0
GPT-5.5	C	268	0	36	0
Claude Sonnet 5	B	304	4 (the four maximum-dose outputs)	0	0
Claude Sonnet 5	C	296	3 (amoxicillin, ceftriaxone, prednisolone school-age: uncapped amounts written in the extraction-only workflow)	8	0
DeepSeek V4 Pro	B	304	3 (two maximum-dose outputs; neonatal rounding record)	0	0

DeepSeek V4 Pro	C	277	0	27	0
Llama 3.1 8B	B	277	27 (Level 1: 2; Level 2: 1; Level 3: 24)	13	14
Llama 3.1 8B	C	270	21 (all Level 3)	16	18

Reading, as the study report states it. Workflow B written-amount deviations among the frontier models number eight (GPT-5.5 1, Claude 4, DeepSeek 3), against seven composite failures (Claude 5, DeepSeek 2). Three-part reconciliation: the six larger maximum-dose outputs are in both counts; the exact-product neonatal amounts of GPT-5.5 and DeepSeek are written-amount deviations only (6 + 2 = 8); Claude's copied neonatal amount is a composite failure only (6 + 1 = 7). Claude's 64.65 mg output is in neither count, being 7.75 percent over. In workflow C a missing amount on a withholding note is correct behaviour (the model was asked to extract, not compute), so GPT-5.5's 36 and DeepSeek's 27 are not failures; Claude computed the amount in 28 of the 36 withholding notes although not asked to; three of those computed amounts exceed the stated maximum by more than 10 percent and a fourth by 7.75 percent. Those three are invisible to Level 4 in workflow C, which recomputes exposure from the mg/kg field, so a deployment that displayed the model's written number without the calculator would not have the protection the workflow-C score implies. For the small model, 14 of 27 missing amounts in B and 18 of 34 in C are on notes that print the amount: the model failed to copy a number that was in the text. The earlier working figure of 28 off for Llama B was computed against the exact product; this table uses the printed reference and supersedes it.

6.12 Stratified results and heat maps (exploratory beyond the pre-specified strata; tools/by_stratum.py; results/by_stratum_run1.csv and .txt). Composite failure rate by complexity, age band, generation method, documentation challenge and drug, for each model and workflow, in long format (stratum kind, stratum, model, workflow, n, primary endpoint rate, composite failure rate). Challenge strata overlap (a record carries several tags) and do not sum to the total.

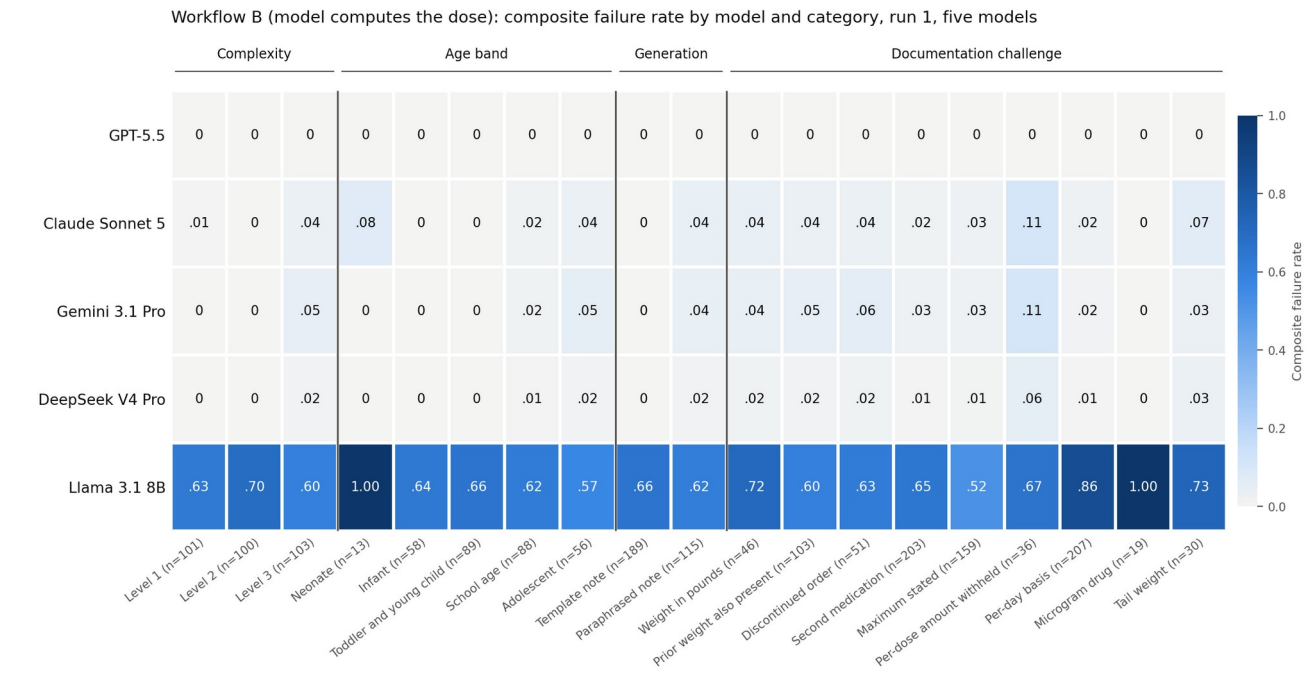


Figure 5. Workflow B composite failure rate by model and category.

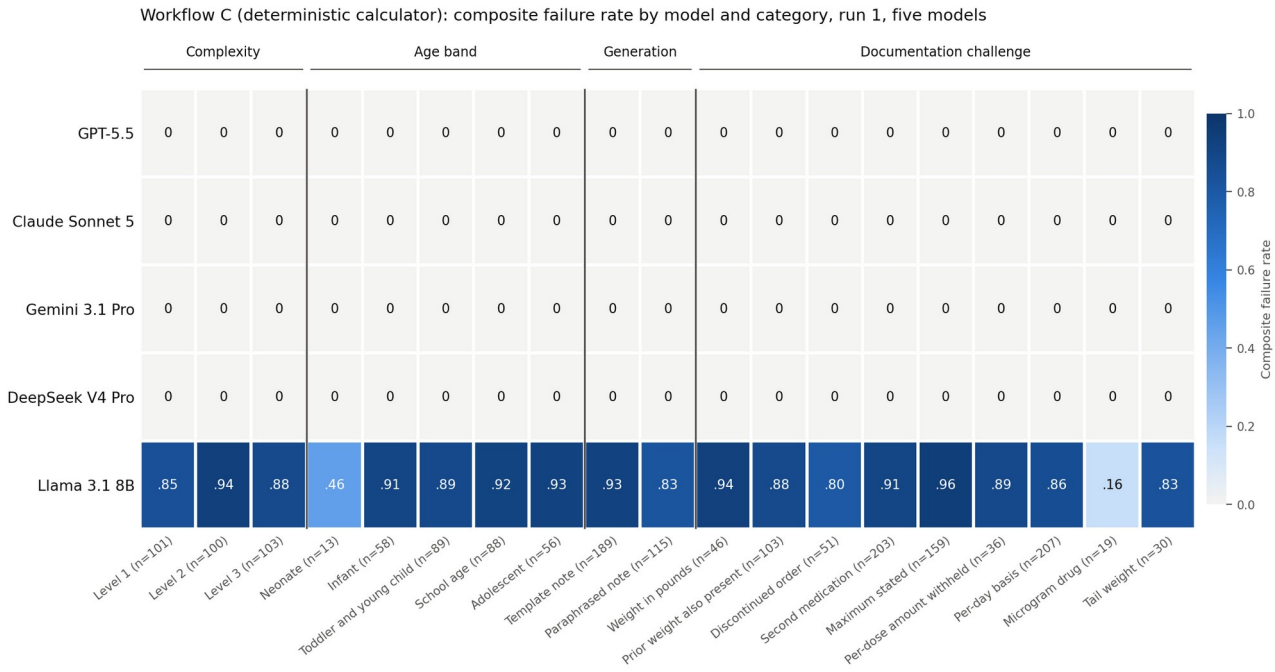


Figure 6. Workflow C composite failure rate by model and category.

Observations, copied from the study report. Frontier models in workflow B: every non-zero cell is accounted for by the seven records of the maximum-dose and neonatal resolutions; the highest cell is "per-dose amount withheld" (Claude 0.111, 4 of 36; DeepSeek 0.056, 2 of 36), where the four maximum-dose notes sit; Claude's neonate cell (0.077) is the single rounding record among 13. Frontier models in workflow C: zero in every stratum. Llama B: fails every per-day-basis order at 0.865 against 0.126 primary-endpoint success, fails all 13 neonatal and all 19 microgram-drug records, and does best on the per-dose fixed-frequency drugs (azithromycin, famotidine, ondansetron, midazolam). Llama C: 0.80 to 0.96 in every stratum except neonate (0.462) and microgram drugs (0.158), where the calculator recovers what the model's own arithmetic lost; these are the only strata in which the calculator helped the small model, and they are small (13 and 19 records). Drug-level rates are in the CSV; frontier drug cells are at ceiling except amoxicillin, ceftriaxone, prednisolone and dexamethasone, for the records above.

6.13 Post hoc maximum check (exploratory; added after external review; tools/max_check.py; results/max_check_run1.csv). Question: would a check of the written per-dose amount against the maximum the model itself extracted have flagged the uncapped outputs? Rule: flag if dose_per_admin_mg exceeds max_per_dose_mg, or dose_per_admin_mg times frequency_per_day exceeds max_per_day_mg, each with a 0.1 percent tolerance, using the model's own extracted maxima; a parallel column uses the scenario's maxima. Non-abstention run1 records, both workflows, 2,432 outputs.

Model	Workflow	Flagged by the model's own maxima	Flagged by scenario maxima
GPT-5.5	B and C	0	0
Claude Sonnet 5	B	5 (the four composite-failure outputs and the 64.65 mg output)	5
Claude Sonnet 5	C	4 (the same notes except	4

		ceftriaxone replaced by the v0 prednisolone note)	
DeepSeek V4 Pro	B	2 (amoxicillin, ceftriaxone)	2
DeepSeek V4 Pro	C	0	0
Llama 3.1 8B	B	0	0
Llama 3.1 8B	C	1 (against its own wrongly extracted maximum; not against the scenario's)	0

Reading. Denominator: 2,432 non-abstention first-run outputs from the four completed models under both workflows, not every output in the study (the repeat runs and the fifth model are excluded). Among the three frontier models the check flags exactly the seven uncapped outputs in workflow B and the four in workflow C (Claude's uncapped amounts written in the extraction-only workflow, which that workflow's Level 4 never uses), and nothing else; agreement between the own-maximum and scenario-maximum columns is complete for them; the single small-model flag is against a maximum the model extracted wrongly. The check is retrospective and has not been tested as a prospective safeguard. The registered exceeds_labeled_max flag, which tests per-day maxima only, misses the amoxicillin per-dose outputs and records the five per-day ones, including the 64.65 mg output that the tolerance passed; the flag does not feed the primary endpoint. The explicit-instruction rerun remains the test of whether the behaviour changes.

6.14 Manual coding (tools/code_errors.py; pending). code_errors.py collects residual extraction errors from the scored files, shuffles them with a fixed seed, hides the model name, shows the note, the output and the automatic codes, and appends each typed code to results/manual_codes.jsonl immediately; quitting and rerunning resumes; the --recode option draws a 10 percent sample with a second seed, presents it again, and writes agreement to results/manual_codes_recode.jsonl. Manual classes from scoring/codebook.md: E1 extraction, E2 normalisation, E3 association, E4 temporal, E5 volume versus dose, E6 other. Status and provenance are in Section 9.

6.15 Export (tools/export_results.py; pending the final run). Writes results/export/ with summary.json (every quoted number), tables as CSV, the run manifest (model ids, dates, settings, prompt hashes, cost), LOCK.json, DEVIATIONS.md and a sanitised configuration.

7. Findings as currently stated, and how they changed under review

7.1 The ceiling result. With four of five models complete, the frontier models performed at or near ceiling on routine extraction (field accuracy 0.996 to 1.000; primary endpoint 0.970 to 1.000), showed no statistically significant pairwise difference after multiple-comparison correction, and exceeded the small open-weight contrast model by 0.64 to 0.94 on the primary endpoint. Run-to-run agreement on the repeated subset was between 0.972 and 1.000 for every completed model. Under review the wording "no two differed" became "no statistically significant pairwise difference was detected, which is not evidence of equivalence", "five pinned models" became "identifiers as used", and "carry beyond this benchmark" became "patterns within the benchmark" (fourth round).

7.2 The three patterns.

First pattern, stale weight. All completed frontier models, in both workflows, missed the same five of the 57 abstention cases in the same way. Each of the five notes, five distinct scenarios, documents only a weight from a

visit three months earlier, and each model used it as the current weight instead of flagging the gap. For the 8-month-old on clindamycin the prior weight is about 20 percent below the reference weight the generator withheld from the note. The prompt gave no rule for how old a documented weight may be. The uniformity across vendors makes a specification gap the most likely explanation; whether an explicit rule changes the behaviour is a separate test the study did not run. Under review (first round) the closing quotation, which had merged this cause with the maximum-dose cause, was split: the stale weight is kept as a specification-gap hypothesis and the maximum failures are named as model behaviour; "not catchable" was judged too absolute and completeness checks were distinguished from range checks.

Second pattern, maximum-dose outputs. Five admission notes gave a weight-based rule, a maximum written into the order, and no per-dose amount, and in each the weight-based product exceeded the maximum. Claude Sonnet 5 wrote the uncapped amount in all five and DeepSeek V4 Pro in two of them, seven outputs in all: 1478 mg of amoxicillin per dose against the order's 875 mg per-dose maximum; 1874 mg of ceftriaxone per administration on a twice-daily order, 3748 mg a day, against a 2 g daily maximum; and 82 mg, 131 mg and 65 mg of prednisolone once daily against a 60 mg daily maximum. Six of the seven overshoot by more than 10 percent and are composite failures; the seventh, 65 mg against 60, overshoots by 8 percent and passed the tolerance, although the separate labelled-maximum flag recorded it. In every one of the seven outputs the model had placed the maximum in the correct structured field before computing the amount without it. GPT-5.5 applied the maximum in all five notes. How this changed under review: in the first round the six outputs were unidentified, and a table with four notes, six outputs, maxima and their fields, label set ids and each model's written amount was added; the ceftriaxone wording was corrected to "1874 mg per administration on a twice-daily order, 3748 mg a day, against a 2 g daily maximum"; in the second round the label claims were found overstated, and the article now calls all four maxima "stated in the order" and names only ceftriaxone's as the label's own, with the amoxicillin 875 mg and prednisolone 60 mg figures documented as the author's derivations; in the third round the claim that "a check catches it today" was found untrue of any check the study had run, so the check was built and run after the fact over all 2,432 outputs, which found the seventh output, 7.75 percent over, that the tolerance had passed, making the count seven outputs over five notes; in the fourth round the maximum check was found stated inconsistently between the article and the report and was made one statement everywhere.

Third pattern, the hybrid workflow. The hybrid design was not the general safeguard it is often assumed to be. Under Section 1 the reduction in composite failure rate had a confidence interval excluding zero for Claude Sonnet 5 only; for DeepSeek V4 Pro it was small and its interval included zero; for GPT-5.5 there was nothing to reduce; under the majority rule the hypothesis is not supported. Under Section 16 two of the four completed models show a gain on the primary endpoint, GPT-5.5's entirely through the nine rounding records, and the verdict on that wording waits for the fifth model. For the small model the calculator raised composite failures from 196 to 271 of 304 and changed their composition from 189 computed exposures more than 10 percent off and 7 not computable to 2 and 269. Under review (first round) the phrase "changed the kind of failure rather than the amount" was found numerically wrong, since the count rose from 196 to 271, and both the count and the composition are now stated; "not a better prompt" was removed from the closing quotation (second round) and the explicit-instruction rerun listed as the next test.

7.3 The small-model reconciliation. The second round asked how 196 composite failures could coexist with 54 written-amount problems. The resolution quotes the scorer path (workflow B scores the model's reported mg/kg fields) and the cross-tab of Section 6.6: 154 records with a correct written amount and a failed composite, of

which 74 carry the milligram figure in the mg/kg field, 48 the per-day value in the per-dose field, 15 a null and 17 other; in workflow C, 233 records with a correct written amount and no computable exposure because the model labelled the order "fixed" and nulled the mg/kg value. The statement now used: the composite is a regimen-reconstruction failure; measured on the amount it wrote, the small model matched the order in about five records of six under both workflows (250 and 249 of 304); measured on the structured regimen a calculator or range check would use, it failed 196 and 271. The article says which number a user of each workflow would see. The third round found that the lede implied the registered measure tests the written amount; the lede and the Figure 4 caption now call it the structured-regimen measure, with the written-amount result beside it, and the fourth round had the legends of Figures 2 and 4 say reported (B) or reconstructed (C) mg/kg exposure, with the written-amount result printed on Figure 4. The written-amount figures 28 and 27, first given without a table, now have the named exploratory tool, rule set, denominator, per-level table and the split of missing amounts by withheld versus printed; the 28 was computed against the exact product and has been superseded by 27 against the printed reference. The eight-versus-seven accounting (third round) is resolved by the three-part reconciliation in Section 6.11.

7.4 The neonatal rounding record. First raised as the seventh frontier error left unexplained (first round), it was traced to one neonatal record whose printed amount is 13 percent from the exact product. The wording "no answer can satisfy both tolerances" was found not established (third round) and restated as "no single internally consistent per-dose amount", after a constructed inconsistent output was run through the locked scorer and passed; the denominators are given (13 percent above the printed amount; 12 percent below the regimen's mg/kg). The record is scored as registered, disclosed, and removed by the 20 percent sensitivity.

7.5 H3 under Section 1 and Section 16. The second round found that H3 has two protocol wordings. Both are now reported side by side with the passages verbatim, paired clustered-bootstrap intervals for each, the reason Section 1 is the analysed definition (the analysis script is one of the seven hashed files, so the analysed definition is dated by the registration hash), and the Section 16 verdict stated as pending the fifth model. The third round asked for "under Section 1" and "under Section 16" throughout, and that the GPT-5.5 effect under Section 16 be identified as a scoring effect of the printed-amount rounding rather than a change in any computed dose, which it now is.

7.6 Hypothesis tally, stated individually. H1 not supported; H2 not supported; H3 not supported under Section 1, pending under Section 16; H4 provisional pending unassisted manual coding; H5 inconclusive. No aggregate count is reported until coding is complete. The third round had found the earlier tally premature; the settled hypotheses are reported as negative results.

7.7 Terminology changes. "Composite reconstruction failure" with its two components reported separately replaced a usage that conflated the composite with the Delphi threshold; "more than 10 percent off" is reserved for computed exposures (first round). "Dose-consequential" is stated as an operational threshold on a constructed intended dose, not a harm claim. "Pinned identifiers" became "identifiers as used", with the DeepSeek caveat, run timestamps and the Ollama digest in the manifest (third round). Development version labels were removed from all public documents; revision history lives in the deviations log and the repository (fourth round).

8. External review history

Four rounds of external adversarial review were processed on 5 October 2026, three on the feature article and its supporting documents and a fourth on the official set. The findings and resolutions below are copied from the review guide; the finding lists from the recorded facts of the journey are consistent with them.

First review round:

Finding	Resolution	Where
Seventh frontier error unexplained	Traced to one neonatal record whose printed amount is 13 percent from the exact product; no output can meet both locked tolerances; scored as registered, disclosed, removed by the 20 percent sensitivity	Report 3.2, 5.11
Six maximum-dose outputs not identified	Six-row table: four notes, six outputs, maxima and their fields, label set ids, each model's written amount; the maximum was extracted into the correct field in all six	Report 5.10; article, second finding
Ceftriaxone maximum wording	"1874 mg per administration on a twice-daily order, 3748 mg a day, against a 2 g daily maximum"	Article; Report 5.10
"Changed the kind of failure rather than the amount"	Count rose from 196 to 271; composition stated	Article, third finding
28 and 27 written-amount figures without a table	Named exploratory tool, rule set, denominator, per-level table, missing split by withheld versus printed	Report 5.12
Closing quote merged two causes; "not catchable" too absolute	Stale weight kept as a specification-gap hypothesis; maximum failures named as model behaviour; completeness versus range checks distinguished	Article, closing quote and third finding
Composite versus Delphi threshold	"Composite reconstruction failure" with two components reported separately; "more than 10 percent off" reserved for computed exposures	Throughout

Second review round:

Finding	Resolution	Where
Small model: 196 composite failures versus 54 written-amount problems	Scorer path quoted: workflow B scores the model's reported mg/kg fields; cross-tab shows 154 records with a correct written amount and a failed composite (74 milligram figure in the mg/kg field, 48 per-day value in the per-dose field, 15 null, 17 other); the article says which number a user of each workflow would see and calls the composite a regimen-reconstruction failure	Report 3.4, 5.6; article, third finding; Figure 2
Label claims overstated	Ceftriaxone 2 g per day is the label's; amoxicillin 875 mg and prednisolone 60 mg are the author's documented derivations; the article calls all four "stated in the order"; no pharmacist review is stated as the standing limitation	Report 5.10, 9
"Except the seven already described"	Eight written-amount deviations and seven	Article; Report 5.12

	composite failures reported as separate counts with the difference named	
H3 has two protocol wordings	Both reported side by side with the passages verbatim, paired clustered-bootstrap intervals for each, the reason Section 1 is the analysed definition, and the Section 16 verdict stated as pending the fifth model	Report 5.2; article, third finding
Figure 2 overflow; figure numbering inconsistent	Figure 2 redrawn; one figure map across all documents	Review guide; Report 12
OSF registration not independently opened	The fact-check packet carries a dated capture of the registration page and the hash outputs; the lock verification is described step by step, including a discrepancy found and resolved on 5 October	Report 4.3
Closing quote: "not a better prompt"	Removed; the explicit-instruction rerun is listed as the next test	Article

Third review round:

Finding	Resolution	Where
Lead implies the registered measure tests the written amount	Lede and Figure 4 call it the structured-regimen measure; the written-amount result stands beside it	Article
"No answer can satisfy both tolerances" not established	Restated as "no single internally consistent per-dose amount"; a constructed inconsistent output was run through the locked scorer and passed; denominators given (13 percent above the printed amount; 12 percent below the regimen's mg/kg)	Report 3.2, 5.11; article
"A check catches it today"	The check was built and run after the fact over all 2,432 outputs against the model's own extracted maxima: it flags the known outputs, nothing else from the frontier models, and one additional Claude output 7.75 percent over a maximum that the 10 percent tolerance had passed; now seven outputs over five notes; quote rewritten	Report 5.10, 5.14; article
H3 framing	"Under Section 1" and "under Section 16" throughout; the analysed definition is dated by the registration hash, since the analysis script is one of the seven hashed files; tally stated individually with H4 provisional and H5 inconclusive	Report 5.2; article
"Pinned identifiers"	"Identifiers as used"; DeepSeek resolves to its current release; run timestamps and the Ollama digest in the manifest	Report 3.3; packet 05
Eight versus seven accounting	Three-part reconciliation	Report 5.12; article
Heat map too dense for the article; editorial note inconsistent	Heat maps moved to the report as supporting material; article carries Figures 1 to 4; one figure map	Article editorial note; review guide
Evidence packet absent	Built from the repository and published with it	Public repository, results/packet

Fourth review round:

Finding	Resolution	Where
Maximum check stated inconsistently between article and report	One statement everywhere: 2,432 eligible first-run outputs of the four completed models; seven end-to-end and four calculator-workflow flags among the three frontier models; one small-model flag against its own wrong maximum; retrospective, not a tested safeguard	Article; Report 5.14
Figure 4 legend and Figure 2 could be read as testing the written amount	Legends now say reported (B) or reconstructed (C) mg/kg exposure; written-amount result printed on Figure 4	Figures 2 and 4
Evidence packet not accessible	Public repository carries results/packet with the raw and scored files it was built from	Public repository
"Carry beyond this benchmark"; "no two differed"; "five pinned models"; architecture reference	Reworded as patterns within the benchmark; no significant pairwise difference detected, not equivalence; identifiers as used; references aligned	Article; Report abstract, 3.3, 5.3
Development version labels in public documents	Removed; revision history lives in the deviations log and the repository	All documents

Reviewer scores given: writing 8.5 of 10, publication readiness 6 of 10 after the second round. A 10 would require no contradictions, an accessible packet, finished or explicitly bounded results, independently checked clinical examples, and language that never turns regimen consistency into a patient wrong-dose rate. Scores after the third and fourth rounds: not recorded.

The questions the author would most value an answer to, as put to the reviewer in the review guide: whether the article now states the small model's result in a way that cannot be read as a wrong-dose rate; whether the handling of the two H3 wordings is transparent enough, given that the verdict differs between them and one is pending; whether the "stated in the order" treatment of the derived maxima is acceptable without a pharmacist read of the five notes, given the stated limitation; whether the three draft quotations are consistent with what the evidence supports, as a basis for the author's approval; and whether, with the heat maps now in supporting material, Figures 3 and 4 carry the piece on their own, and the seventh, within-tolerance output is stated at the right weight.

9. Integrity and provenance

9.1 Integrity controls (from the system architecture document).

Control	Mechanism
Test set immutability	SHA-256 in LOCK.json, published on OSF; runner and scorer refuse on mismatch
Scorer, calculator, prompt, schema and registered-analysis immutability	One combined hash at lock; the scorer recomputes it on every run; any change requires a logged deviation; a working-copy edit to the analysis script on 5 October was caught by this check, reverted, and moved outside the hashed set
Model pinning	Exact identifiers in config/models.yaml and in each raw file's _config line
Generator and evaluated model separation	Family check in harness/common.py refuses overlap
Resumability	Done set keyed on parsed output; last attempt wins at scoring; identical for all models
Blinding of manual coding	Model name hidden; randomised order; fixed seed; append-only file; unassisted
Secrets	.env git-ignored; secret_scan.py before any push, including

	results/logs; keys rotated after the study
Budget	200 USD ceiling in config; spend recorded per provider in the manifest
Deviations	protocol/DEVIATIONS.md, one dated row per change with rationale

9.2 Lock verification narrative. Verification was performed on 5 October 2026. The three data files recompute to the registered values with shasum: records 492d7ca2209719eb8bf96548403dab1e26cb9fed36be3482733dd69790cea7f4; scenarios 432be74fe65c2164ad9e21705821f5814055994f389526614aa744a671c220cb; repeat subset f1012bdcb01f9cdcf80cf000b0ad6831352d9a98a6e57a0e8ebd207dbdfa4196. scoring/score.py was last changed in commit 7c948c4 (3 October, before the lock) and git reports the working file identical to that commit (blob ace4c0fa); no commit after the lock commit 528c2b5 touched any of the seven hashed files. The first recomputation of hash_scoring() on 5 October returned a different value from the registered one, because the malformed-row correction to analysis/analyze.py (the deviation that counts malformed non-abstention rows as failures) had been applied in the working copy of a hashed file. The locked analyze.py was restored byte for byte from the repository; the correction was moved to analysis/analyze_malformed_as_failure.py outside the hashed set, with its diff saved as results/logs/analyze_malformed_as_failure.diff (15 lines); hash_scoring() then returned the registered value 7be9bc8704e09b94014741847f16b047b115c7b2c7c7846ec01afe03445461ed. The locked and corrected scripts produce identical run1 tables (results/analysis_primary_locked.txt and results/analysis_primary_corrected.txt differ only in the print order of one line). All scoring to date was performed before the working-copy change, under the registered scorer; the scorer's refusal rule is what surfaced the discrepancy. The evidence packet (01_lock_verification.txt) carries LOCK.json, the shasum output for the three data files, the hash_scoring() output, the empty list of post-lock commits to hashed files, and HEAD. The prepared answer to the challenge that the scorer hash does not match a shasum of score.py: the lock's scoring hash covers seven files via hash_scoring(); git shows none changed by any commit since the lock; the working-copy edit was caught by the recomputation, reverted, and moved outside the hashed set; the registered value is reproduced and the packet carries the command and output.

9.3 Deviations and clarifications after registration (all in protocol/DEVIATIONS.md), copied from the study report:

Date	Item	Why
4 Oct	Workflow A (rules baseline) not run	Outside H1 to H5; calculator in C is the non-LLM arithmetic reference; decided before results
4 Oct	Resumed runs re-query records with no parsed output; scorer keeps the last attempt	Runner's documented behaviour; identical for all models; before and after malformed counts reported
4 Oct	Gemini requests split across two Google Cloud projects	Per-project daily cap; same model, endpoint, parameters
4 Oct	Two DeepSeek run2 records persistently empty, scored malformed	Protocol rule; three attempts
5 Oct	H3 wording: Section 1 (dose-consequential rate) versus Section 16 (primary endpoint); script follows Section 1	Both differences reported
5 Oct	sources/age_bands.json verification note replaced by the band mapping statement; no boundaries changed	Housekeeping; mapping stated in the paper
5 Oct	Malformed non-abstention rows count as failures on primary endpoint, complete	Definition alignment with Section 12; locked file set unchanged; run1 unaffected

	regimen and dosing consistency; previously dropped from the bootstrap denominator. First applied inside the locked analyze.py; the locked file was then restored and the correction implemented as analysis/analyze_malformed_as_failure.py outside the hashed set, diff retained; outputs identical on run1 (no malformed rows); the corrected script is used for any file containing malformed rows (two DeepSeek run2 records)	
5 Oct	"Incorrect inference" abstention category not scored separately; folded into unsupported-value hallucination	Scorer locked; stated as implemented
5 Oct	Pairwise Holm-corrected contrasts implemented in analysis/pairwise.py after registration; mixed-effects regression not performed (degenerate at ceiling)	Analysis code not under lock
5 Oct	Manual coding: first 40-item pass with assistant input discarded; unassisted recoding	Provenance
5 Oct	Clarification: dose_consequential is a composite (exposure more than 10 percent off, or not computable); components reported separately; exceeds_labeled_max tests per-day maxima only	Scorer locked; stated as implemented
5 Oct	Clarification: one test-set record (neonatal dexamethasone) has a printed amount 13 percent from the exact product, so Level 2 and Level 4 cannot both be met in workflow B; scored as registered, disclosed, removed by the 20 percent sensitivity	Test set locked; scorer locked
5 Oct	Exploratory written per-dose amount analysis added (tools/admin_amount.py)	Not pre-specified; reported as exploratory with its rule set
5 Oct	Exploratory stratified table and heat maps by challenge tag, generation method and drug added (tools/by_stratum.py); complexity and age band were pre-specified moderators, the rest were not	Not pre-specified; reported as exploratory
5 Oct	H3 reported under both protocol wordings (Section 1 composite rate; Section 16 primary endpoint); Section 1 kept as the analysed definition, reason stated	Protocol names two outcome measures for one hypothesis
5 Oct	Clarification: workflow B Level 4 scores the model's reported mg/kg values, not a value derived from its written amount; the written amount is scored at Level 2 and in the exploratory check	Scorer locked; stated as implemented; earlier report wording corrected
5 Oct	Clarification: the lock's scoring hash is hash_scoring() over seven files (scorer, calculator, three prompt files, output schema, analysis script), not SHA-256 of score.py alone; recomputed and matching after the restoration above; earlier report wording corrected	Documentation
5 Oct	Exploratory post hoc maximum check added (tools/max_check.py); found a seventh uncapped output inside the 10 percent tolerance	Not pre-specified; reported as exploratory

5 Oct	Neonatal record: probe of a constructed output through the locked scorer; an earlier draft's claim that no output could pass both levels withdrawn and restated	Scorer locked; probe files removed; scored files unchanged
5 Oct	Model identifiers described as used, with run timestamps and the local model digest in the manifest; deepseek-v4-pro resolves to the provider's current release	Provider documentation

The 2 October both-bases rule change (gentamicin carrying both per-dose and per-day rows, downgraded to a warning, commit b249a75) is also logged in DEVIATIONS.md; it predates the registration.

9.4 Manual-coding provenance. A first 40-item coding pass carried assistant input (one suggested code and one suggested reclassification) and was discarded; the file is retained as results/manual_codes_assisted_pass_discarded.jsonl. The author codes all 288 residual errors, plus Gemini's once its runs complete, unassisted, from scoring/codebook.md, with tools/code_errors.py (model name hidden, randomised order with a fixed seed, append-only file); a random 10 percent is re-coded after two weeks with --recode 0.10 and intra-rater agreement is reported on the unassisted file. No assistant suggests a code for any item. The prepared answer to "Was the error coding human?": a first assisted pass was discarded; the reported coding is unassisted and the re-code is intra-rater on that file.

9.5 Model identifiers as used. gpt-5.5-2026-04-23 (the identifier carries a release date); claude-sonnet-5 and gemini-3.1-pro-preview (the identifiers name a release; Gemini on an OpenAI-compatible endpoint); deepseek-v4-pro, which resolves to the provider's current release at call time and is therefore not immutable; llama3.1:8b on Ollama, local, digest 46e0c10c039e. Because of the DeepSeek caveat the manifest records first and last call timestamps per run file (all test runs 4 to 5 October 2026) rather than claiming every provider version was fixed. The manifest (packet 05_manifest/) holds per-file model, endpoint, workflow, run, attempts, first and last timestamps and prompt hash; the Ollama digest; and models.yaml with environment-variable names only. Gemini 3.1 Pro is a preview model: identifier, manifest and raw outputs are released and the limitation is stated; preview identifiers may be withdrawn by vendors. The generator and auditor model, qwen2.5:14b (local), belongs to a family distinct from every evaluated model, enforced by the family check in harness/common.py; a development assistant family that is also evaluated is disclosed and was evaluated only through its pinned API.

9.6 Source provenance. sources/dosing_sources.csv with label set ids (the three maximum-dose labels: amoxicillin 06e5c07d-d95c-4955-8d35-a4703ca2359e, ceftriaxone 53df951b-2eee-44b1-bae2-df7ed85bc0ed, prednisolone 286a6032-41de-435a-b6e6-14052b66b85d); growth file download record; age_bands.json. Manuscripts previously drafted by an external writer on overlapping topics were audited and set aside because their data provenance could not be established; nothing from them is used.

9.7 Known limitations of the implementation (from the system architecture document). A single author operates every stage; the realism auditor and paraphrase generator are the same model family; the prompt omits a staleness rule for documented weights (now a reported finding); workflow B's Level 4 trusts the model's reported mg/kg values, so a model that copies the printed amount correctly can still fail on its own inconsistent mg/kg fields, which is why the written amount is reported separately; the per-day maximum flag does not test per-dose maxima; one test-set record has a printed amount more than 10 percent from the exact product; preview model identifiers may be withdrawn by vendors; local inference for Llama is deterministic at the chosen

settings, so its stability figure is not comparable to API models; resumed runs give unparsed outputs an extra attempt, applied uniformly and reported.

10. Documents produced

10.1 The official document set, issued on 5 October 2026 without development version labels.

Document	Purpose and audience
From Fields to Dose: Feature Article	The publishable text for an outlet with editorial control; four sections, one author mention, four figures (Figures 1 to 4), a sources list, three draft quotations awaiting approval, editorial notes to be removed before publication; audience: the outlet's editors and readers
From Fields to Dose: Study Report	The single self-contained account of the study that every number in the article traces to; audience: the author, reviewers and fact-checkers
From Fields to Dose: System Architecture	Software, data model, pipeline, integrity controls, operations and release plan; audience: anyone reproducing the lock, scorer and analysis in code
From Fields to Dose: Review Guide	Contents of the review set, review history, pending items, standing constraints, and the questions the author would most value an answer to; audience: external reviewers
From Fields to Dose: Supporting Information	Supporting material for the outlet; contents not recorded in the sources beyond its name
Protocol v1.0 (OSF)	The registered design, whose title is kept as registered; audience: OSF readers and reviewers verifying the pre-registration

10.2 What each contains. The feature article: a lede on the pre-registered test, then "What the work is" (the study, the registration, the synthetic test set, the five models and two workflows, with Figures 1 and 2), "What came before" (the pediatric medication error literature and the two comparators), "How the study was assessed" (TRIPOD-LLM, NICHD age bands, WHO and CDC references, the two choices stated openly), "Current stage" (research in progress, the ceiling result with Figure 3, the rounding shortfall, the three patterns, Figure 4, the written-amount check, the hypothesis tally, the closing quotation), ten sources for verification, and editorial notes naming the three Gemini insertion points. The study report: abstract; the question and why it matters; literature and standards; study design (constraints, test set, models and workflows, scoring, error taxonomy, statistics); validation before lock (development phase, audits, lock); results to date in fourteen subsections; pending work and why each item stays in; deviations and clarifications; decisions and rationale; anticipated challenges and prepared answers; publication and evidence plan; evidence packet; figures; document map. The system architecture document: design principles; repository layout; data model (scenario, record, raw output, scored row, manual codes, supplementary tables); pipeline (sources, generation, lock, execution, scoring, analysis, manual coding, export); integrity controls; operations (standard sequence BLOCKS 0 to 8, running and resuming, provider quotas, per-call cost, verification commands); public release (what is public, what is not, procedure); known limitations. The review guide: what the package is; contents table; evidence packet table; the four review-round tables; what is still pending; standing constraints; the five questions for the reviewer.

10.3 Where the figures appear. Figure numbers are the same in the article, the report and the evidence packet. The article carries Figures 1 to 4; Figures 5 and 6 are report-only supporting material available to the outlet. Figures are supplied as separate PNG files at 200 dpi with the data file behind each.

Figure	File	Data source	Appears in

1 Study pipeline	F2D_Figure_Study_Pipeline.png	Protocol v1.0; LOCK.json	Article, report
2 Two workflows	F2D_Figure_Two_Workflows.png	Protocol Section 12; harness configuration	Article, report (redrawn after the second round)
3 Primary endpoint	F2D_Figure_Primary_Endpoint.png	analysis_primary_four_models.txt	Article, report; Gemini row filled on completion
4 Small model failure composition	F2D_Figure_Small_Model_Failure_Composition.png	scored run1 files, Llama B and C	Article, report; legend says reported (B) or reconstructed (C) mg/kg exposure; written-amount result printed on the figure
5 Heat map, workflow B	F2D_Figure_Heatmap_Workflow_B.png	by_stratum_run1.csv	Report only; Gemini row filled on completion
6 Heat map, workflow C	F2D_Figure_Heatmap_Workflow_C.png	by_stratum_run1.csv	Report only; Gemini row filled on completion

10.4 The review set as issued. Fields_to_Dose_Feature_Article.docx and .pdf; Fields_to_Dose_Study_Report.docx and .pdf; Fields_to_Dose_System_Architecture.docx and .pdf; the six figures under Figures/; the review guide. The evidence packet, the code, the locked data and every raw and scored output are in the public repository.

10.5 Evidence packet contents (results/packet in the public repository, built by script from the working repository, 31 files, commit 8776fe6).

Part	Contents
01_lock_verification.txt	LOCK.json, shasum of the three data files, hash_scoring() output, empty post-lock commit list for the hashed files, HEAD
02_row_level/	Note, scenario, every model's output and scored row in both workflows for the five stale-weight records, the five maximum-dose notes and the neonatal record (stale_weight, maximum_dose, neonatal_rounding; as text and JSONL)
03_written_amount/	admin_amount, by_stratum, per_drug and max_check CSVs, their scripts, the written-amount versus composite cross-tab
04_h3/	Locked and corrected primary analyses, pairwise contrasts, H3 under both protocol wordings
05_manifest/	Per-file model, endpoint, workflow, run, attempts, first and last timestamps, prompt hash; Ollama digest; models.yaml (environment-variable names only)
06_deviations/	DEVIATIONS.md, LOCK_hashes.txt

Still to add with the final version, as listed in the study report: final analysis tables with model, workflow, numerator, denominator, CI and adjustment method for every quoted number (analysis_primary.txt, analysis_pairwise.txt, analysis_stability.txt, per_drug_run1.csv, admin_amount_run1.csv, by_stratum_run1.csv); the hypothesis ledger with the analysis version; the five stale-weight records with scenario ids, note text, each model's weight output per workflow and the generator's withheld reference weight; the maximum-dose notes with note text, scenario fields, all outputs with parsed JSON, scored rows and label set identifiers; the neonatal rounding record with note text, three models' outputs, scored rows and the scorer's Level 4 path; calculator traces for two workflow-B and two workflow-C failures, note to extracted fields to calculator input to result to scored outcome, with "no exposure computable" separated from "exposure more than 10 percent off"; the rounding audit listing the 53 records and the one record above 10 percent; registration evidence (dated capture of the OSF registration page showing the hashes, LOCK.json, LOCK_hashes.txt, the shasum and hash_scoring()

outputs, the git log of score.py); source provenance (dosing_sources.csv with label set ids, growth file download record, age_bands.json); figures as PNG at 200 dpi with the data file behind each; approved quotes with the author's written approval.

10.6 Publication plan. Paper: Results and Discussion drafted from the four-model analysis, completed with Gemini; TRIPOD-LLM checklist; venue after results are final. Article: four-part format with one author mention, four figures, three Gemini insertion points (the ceiling comparison in the first paragraph of Current stage, the stale-weight finding, and the workflow comparison), sources list, quotes awaiting approval; placement must be in an outlet with editorial control, a byline that is not the author's, and no press-release label. Registration, project, repository and ORCID record form the verification chain. No independent commentator has been quoted; the outlet may add one.

11. Operating procedures still to run

The commands below are copied exactly from the operating record. The standard sequence is SETUP_GUIDE BLOCKS 0 to 8: 0 environment and keys; 1 tests and smoke; 2 sources; 3 development corpus and runs; 4 test corpus, audits, self-audit, lock; 5 protocol finalisation and OSF registration; 6 test runs; 7 scoring, analysis, stability, manual coding; 8 export and hand-off. The project stands in BLOCK 6 for Gemini and BLOCK 7 for the rest.

11.1 Gemini chain (one project; rerun after the key swap for the second project). Runs the workflow B run1 remainder, then run2 and run3 for workflows C and B on the repeat subset; the runner resumes from existing parsed output.

```
PYTHONUNBUFFERED=1 caffeinate -i nohup zsh -c 'python -m harness.runner --split test --workflow B --
run-id run1 --model gemini-3.1-pro-preview; for w in C B; do for r in run2 run3; do python -m
harness.runner --split test --workflow $w --run-id $r --subset data/test/repeat_subset.txt --model
gemini-3.1-pro-preview; done; done' > results/logs/gemini_evening1.log 2>&1 &

sleep 60; tail -n 3 results/logs/gemini_evening1.log
```

11.2 Status of runs (counts records with parsed output; target 361 for run1 files, 108 for repeat files).

```
python3 - <<'PY'

import json,glob

for f in sorted(glob.glob("results/raw_test_*_run1.jsonl"))+sorted(glob.glob("results/
raw_test_*_run[23].jsonl")):

    ok=set()

    for l in open(f):

        o=json.loads(l)

        if "record_id" in o and o.get("parsed") is not None: ok.add(o["record_id"])

    target=361 if "run1" in f else 108

    print(f"{len(ok):4d} / {target}  {'DONE' if len(ok)>=target else '':5s} {f.split('/')[-1]}")

PY

pgrep -f harness.runner >/dev/null && echo "runner: RUNNING" || echo "runner: STOPPED"
```

11.3 Key swap between the two Google Cloud projects (both keys kept in .env; one active). Run when the active project's daily quota is exhausted, then rerun the Gemini chain.

```
python3 - <<'PY'

L=open(".env").read().splitlines(); d={l.split("=",1)[0]:i for i,l in enumerate(L) if "=" in l}

a,b=d["GOOGLE_API_KEY"],d["GOOGLE_API_KEY_PROJECT1"]; va=L[a].split("=",1)[1]; vb=L[b].split("=",1)
[1]

L[a]="GOOGLE_API_KEY="+vb; L[b]="GOOGLE_API_KEY_PROJECT1="+va; open(".env","w").write("\n".join(L)
+"\n"); print("swapped; active key now ends with",vb[-4:])

PY
```

11.4 Final scoring and analysis (after all runs complete; expects 30 scored files).

```
python -m scoring.score --raw results/raw_test_*.jsonl 2>&1 | tail -n 2; ls
results/scored_test_*.jsonl | wc -l

python -m analysis.analyze --scored 'results/scored_test_*_run1.jsonl' | tee
results/analysis_primary.txt > /dev/null

rm -f results/analysis_stability.txt; for m in gpt-5.5-2026-04-23 claude-sonnet-5 gemini-3.1-pro-
preview deepseek-v4-pro llama3.1_8b; do for w in C B; do echo "## $m wf=$w" >>
results/analysis_stability.txt; python -m analysis.stability --scored results/scored_test_${w}_$
{m}_run1.jsonl results/scored_test_${w}_${m}_run2.jsonl results/scored_test_${w}_${m}_run3.jsonl >>
results/analysis_stability.txt 2>&1; done; done

python -m analysis.pairwise --scored 'results/scored_test_*_run1.jsonl' | tee
results/analysis_pairwise.txt

python -m tools.per_drug --scored 'results/scored_test_*_run1.jsonl' > results/per_drug_run1.txt

python -m tools.admin_amount --scored 'results/scored_test_*_run1.jsonl' | tee
results/admin_amount_run1.txt

python -m tools.by_stratum --scored 'results/scored_test_*_run1.jsonl' | tee
results/by_stratum_run1.txt

python -m tools.max_check

python -m tools.export_results 2>&1 | tail -n 5; ls results/export
```

Note for the author: any scored file containing malformed rows (the two DeepSeek workflow B run2 records) is analysed with analysis/analyze_malformed_as_failure.py, the logged deviation; the command for that script is not recorded in the operating record. Figures 3, 5 and 6 are regenerated with the Gemini row after this block; the figure-generation command is not recorded.

11.5 Manual coding (unassisted; the author alone).

```
python -m tools.code_errors --scored 'results/scored_test_*_run1.jsonl'
```

Re-code after two weeks: the same command with --recode 0.10.

```
python -m tools.code_errors --scored 'results/scored_test_*_run1.jsonl' --recode 0.10
```

11.6 Lock verification (repeat before every release and after the final scoring). Expected: the three shasum values and the hash_scoring() value equal the four hashes in Section 4.4; the git log is empty.

```
shasum -a 256 data/test/records.jsonl data/test/scenarios.jsonl data/test/repeat_subset.txt

python3 -c "from tools.lock_testset import hash_scoring; print(hash_scoring())"

git log --format='%h %ad %s' --date=iso 528c2b5..HEAD -- scoring/score.py generator/calculator.py
harness/prompts/system.txt harness/prompts/task_B.txt harness/prompts/task_C.txt
harness/output_schema.json analysis/analyze.py
```

11.7 Public release. tools/make_public_release.py builds ../from-fields-to-dose-public, writes README, LICENSE (MIT), LICENSE-DATA (CC BY 4.0), CITATION.cff and .gitignore, and runs a secret-pattern scan; then, in that directory, the repository is initialised and pushed. Repeat at each milestone (fifth model complete; coding complete; acceptance); tag the commit and record the tag in the paper.

```
python -m tools.make_public_release

git init -q; git add -A; git commit -q -m "..."; git branch -M main; git remote add origin
https://github.com/VinodRufus/from-fields-to-dose.git; git push -u origin main
```

What the release contains: README, licences and CITATION.cff (carrying ORCID and the OSF DOI); config/models.yaml (environment-variable names only); sources/; generator/, harness/ (prompts, schema, runner, client), scoring/, analysis/, tools/, tests/; data/test/ (locked split, LOCK.json, repeat subset, audits) and data/dev/; protocol/ (Protocol v1.0 docx and PDF, LOCK_hashes.txt, DEVIATIONS.md, README) without the archive; results/ raw and scored run files, analysis outputs, supplementary CSVs, manual coding files and the evidence packet, without logs and backups; docs/ with the official documents and figures; SETUP_GUIDE.md. Excluded: .env and any environment file, the virtual environment, caches, .DS_Store, results/logs, results/backup, protocol/archive, and any file not on the allow-list. The release directory has its own git history with no connection to the working repository's history. At acceptance: update the OSF project with the paper DOI and the repository URL; rotate all provider keys.

12. Pending work and release checklist

Item	Detail	Owner	Status
Gemini 3.1 Pro workflow B run1 remainder	19 records (342 of 361 complete); next quota reset; Gemini chain, Section 11.1	author	pending
Gemini 3.1 Pro run2 and run3, both workflows	108-record repeat subset; about 450 calls in total with the run1 remainder, split across the two Google Cloud projects; key swap, Section 11.3	author	pending
Score all 30 run files	Section 11.4, first command; expects 30 scored files	author	pending
Rerun analyze, stability, pairwise, per-drug, admin_amount, by_stratum, max_check	Section 11.4; five models	author	pending
Malformed-row analysis	analysis/ analyze_malformed_as_failure. py for files with malformed rows (two DeepSeek run2 records)	author	pending
Regenerate Figures 3, 5 and 6	Data sources:	author	pending

with the Gemini row	analysis_primary.txt, by_stratum_run1.csv		
H3 under both wordings for five models	Section 1 (composite rate, analyze.py) and Section 16 (primary endpoint, pairwise.py)	author	pending
Insert Gemini results in the article at three points	Ceiling comparison in the first paragraph of Current stage; stale-weight finding; workflow comparison; sentences supplied from the same analysis script	author	pending
Add Gemini rows to the study report	Primary table, H3 tables, abstention, stability, written amount, maximum check	author	pending
Unassisted manual coding	288 residual errors plus Gemini's; tools/code_errors.py; Section 11.5	author	pending
10 percent re-code after two weeks	--recode 0.10; intra-rater agreement reported on the unassisted file	author	pending
Settle H4 and state the aggregate hypothesis tally	After unassisted coding	author	pending
Author's written approval of the three draft quotations	In the article	author	pending
TRIPOD-LLM checklist	Released with the paper	author	pending
Dated OSF registration capture	Added to the evidence packet (registration evidence)	author	pending
Export	tools/export_results.py: summary.json, tables, run manifest with final provider-dashboard cost, LOCK.json, DEVIATIONS.md, sanitised configuration	author	pending
Refresh the evidence packet from the final export	Including the items "still to add" in Section 10.5	author	pending
Lock verification after the final scoring	Section 11.6; expected hashes in Section 4.4	author	pending
Public release repository regenerated and tagged	Section 11.7; at each milestone (fifth model complete; coding complete; acceptance); tag recorded in the paper	author	pending
Paper Results and Discussion completed with Gemini	Venue after results are final	author	pending
Exploratory follow-up (a)	Five stale-weight records rerun with an explicit staleness rule in the prompt, all models, both workflows, about 50 calls; post hoc	author	pending
Exploratory follow-up (b)	Level 2 sensitivity scoring the per-dose field against the exact product instead of the printed amount; post hoc	author	pending
Exploratory follow-up (c)	Five maximum-dose notes rerun with an explicit instruction to apply any stated maximum, all models, workflow B, about 25 calls; post hoc	author	pending
OSF project update at acceptance	Paper DOI and repository URL	author	pending

Provider key rotation at acceptance	All provider keys	author	pending

13. Decisions and rationale; anticipated challenges and prepared answers

13.1 Decisions and rationale (copied from the study report).

Decision	Rationale
Solo authorship, no clinician or pharmacist	Ground truth by construction; gates and audits; blinded self-audit reported; limitation stated
Synthetic data only	Releasability; no protected information; controlled failure paths; generalisation limitation stated
Pre-registration with hashes before any test run	Removes post hoc flexibility; later changes are logged deviations
21 drugs from FDA labeling, one label each	Verifiable provenance; ranges drive plausibility gates
361 records	Audit removals and two-variant design; registration states 361
Repeat subset 108, three runs	Stability at affordable cost
Gemini kept; two projects	Pre-registered model; quota is per project
Grok not added	Not pre-registered; no expected change in conclusions
Locked scorer applied as registered on the neonatal rounding record	Changing a locked scorer after results is the deviation a reviewer reads as tuning; disclosure plus the pre-registered sensitivity is the defensible route
Public-document standard	ORCID on every document; no clock times; no em or en dashes; human authorial voice; reviewer-first drafting
AI assistance	Protocol Section 20: assistants as development and drafting aids under the author's direction; not evaluated models; overlapping family evaluated only through its pinned API; assisted coding pass discarded and disclosed

13.2 Anticipated challenges and prepared answers (copied from the study report).

Challenge	Answer
Synthetic notes are easier than real ones	Agreed and stated; the benchmark bounds the floor; discriminating strata were pre-identified
No clinician validated the gold standard	Ground truth by construction from labelled dosing; committed gates and audits; blinded self-audit rate reported
Frontier models are at ceiling, so the study shows nothing	Predicted in the protocol; the study found a uniform abstention failure, six maximum-dose outputs with the maximum correctly extracted, and a workflow interaction that reverses for a weak extractor; four hypotheses failed and are reported
Things changed after registration	Every change is in DEVIATIONS.md with date and reason; test-set and scorer hashes match the registration
The printed per-dose amount and the mg/kg regimen disagree in some notes	Yes: about 53 of 304 by more than 1 percent, one by more than 10 percent; in workflow B the rounding reaches Level 4 when a model copies the printed amount, which happened once; the record is traced (5.11), scored as registered, and removed by the 20 percent sensitivity
The stale-weight result is a prompt artefact	By design the prompt had no staleness rule; the finding is that models do not supply one; a rule test is a listed follow-up, not a claim
The maximum-dose result is a prompt artefact	The maximum was in the note in plain words and the model wrote it into the correct field; the prompt's example object includes the maximum fields; an explicit-instruction rerun is a listed follow-up
The amoxicillin and prednisolone maxima are not label maxima	Correct, and stated (5.10): they are documented derivations recorded in dosing_sources.csv; the model's task was to apply the maximum written in the order it read; the article calls them maxima stated in the order

The small model's "wrong dose" rate is overstated	The composite is a regimen-reconstruction measure on the reported or reconstructed mg/kg values; the written amount is scored separately and was within 10 percent in 250 and 249 of 304; both are reported with the cross-tab (5.6) and the article says which number a user of each workflow would see
H3 was analysed on the wording that fails	Both wordings are reported side by side with the protocol passages (5.2); Section 1 names the hypothesis's quantity; under Section 16 the verdict is pending the fifth model and the GPT-5.5 effect is a rounding artefact
The tolerance hides a cap violation	Yes: one output 7.75 percent over a stated maximum passed Level 4; the registered per-day flag recorded it (exceeds_labeled_max True) but the flag does not feed the primary endpoint; reported (5.10, 5.14)
The scorer hash does not match a shasum of score.py	The lock's scoring hash covers seven files via hash_scoring(); git shows none changed by any commit since the lock; a working-copy edit to the analysis script was caught by the recomputation, reverted, and moved outside the hashed set; the registered value is reproduced and the packet carries the command and output (4.3)
The hybrid workflow is supposed to be safer	It met the criterion for one model and reversed for the small model; it cannot repair a wrong dose basis; it also masks a written amount that exceeds the maximum (5.12)
"Dose-consequential" implies harm	It is an operational threshold on a constructed intended dose; the composite's two components are reported separately and the Delphi figure is applied only to computed exposures
Were models given multiple attempts?	One repair per call; resumed runs re-queried unparsed outputs identically for all models; counts reported
Gemini is a preview model	Pinned identifier, manifest and raw outputs released; limitation stated
A development assistant family is also evaluated	Disclosed; evaluated only via pinned API; generator and auditor are a different family
Was the error coding human?	A first assisted pass was discarded; the reported coding is unassisted and the re-code is intra-rater on that file
Who is the independent commentator?	None fabricated; the assessment section rests on published standards; the outlet may add one

The section references in parentheses in both tables (5.11, 5.10, 5.6, 5.2, 5.14, 5.12, 4.3) are the study report's own section numbers, kept as copied.

Final results with the fifth model

Gemini 3.1 Pro completed all runs on 6 October (run 1 on 361 records in both workflows; three runs on the 108-record subset; zero errors in the final chains), across two Google Cloud projects under one billing account after the per-project daily quota. The final scoring and analysis were run on the full set of 30 files. What changed and what did not:

Item	Final value
Gemini 3.1 Pro, end-to-end	field accuracy 1.000; complete regimen 0.997; composite failure 0.016 (5 of 304); primary endpoint 0.980 (0.964 to 0.993)
Gemini 3.1 Pro, hybrid	all 1.000; composite failure 0
Gemini stability	1.000 (hybrid), 0.991 (end-to-end)
Pairwise	no significant difference among the four frontier models in either workflow (largest adjusted p 0.92 is the smallest); every frontier model exceeds Llama by 0.641 to 0.661 (end-to-end) and 0.941 to 0.944 (hybrid)
Abstention	all four frontier models miss the same five stale-weight records in

	both workflows, writing the prior weight as current
Maximum-dose outputs	nine over five notes (Claude 5, DeepSeek 2, Gemini 2); seven beyond the tolerance; amoxicillin per-dose maximum missed by three of four frontier models; GPT-5.5 applied the maximum in all five
New: reporting-convention finding	Gemini capped the written amount correctly in four notes and reported the order's nominal mg/kg rather than the capped figure; the prompt does not state the convention; scored as registered (four composite failures with correct written amounts) and disclosed as a clarification
H3 under Section 1 (composite rate)	Claude -0.016, Gemini -0.016 supported; GPT-5.5 null; DeepSeek -0.007 not; Llama +0.247 reversed; 2 of 5, not supported
H3 under Section 16 (primary endpoint)	GPT-5.5 +0.030, Claude +0.016, Gemini +0.020 supported; DeepSeek +0.010 not; Llama -0.273 reversed; 3 of 5, supported
Frontier composite failures, end-to-end	twelve: Claude 5, DeepSeek 2, Gemini 5; all in paraphrased notes
Written-amount deviations, end-to-end	ten: GPT-5.5 1, Claude 4, DeepSeek 3, Gemini 2
Maximum check denominator	3,040 eligible first-run outputs; nine end-to-end and four hybrid flags among the frontier models; one small-model flag against its own wrong maximum
Figures	3, 5 and 6 regenerated with the Gemini row

Still pending: unassisted manual coding and the re-code; TRIPOD-LLM checklist; dated registration capture; approval of any quotation; public repository refresh from the final export; key rotation at acceptance.

Repository

Code, synthetic data, locked test split, protocol, raw and scored outputs, analysis outputs, evidence packet, figures and documents: https://github.com/VinodRufus/from-fields-to-dose